Tektronix Auxiliary Front Panel (AFP)

AFP Hardware Reference Manual — Rev B

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Document Revision: 2.0 — AFP Hardware Reference Manual

Issued: May 18, 2026

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1. System Overview

The Tektronix Auxiliary Front Panel (AFP) Rev B is an external auxiliary control surface for Tektronix MSO oscilloscopes. The hardware provides physical knobs, push buttons, rotary encoders, and LED indicators to support common and extended oscilloscope control functions from an external front-panel-style device.

This document focuses on the Rev B hardware design, including the PCB architecture, power distribution, processor connections, I/O expansion, LED driver system, user controls, assembly, bring-up, and verification. Software and oscilloscope command handling are discussed only where needed to explain the hardware interfaces.

1.1 Block Architecture

Two processors handle different responsibilities, linked by a 3.3 V UART:

- **Raspberry Pi 4 — high-level system controller**
  Runs Linux and the main AFP application. Handles VISA/SCPI communication with the oscilloscope through the Pi 4's USB or Ethernet interfaces. It also manages the external display connection, macro storage, and oscilloscope state tracking.
- **Raspberry Pi Pico — real-time front-panel I/O controller**
  The Pico/RP2040 module is mounted on the AFP PCB. It reads buttons and encoders through four MCP23017 I^2C GPIO expanders, drives the LED indicators through three daisy-chained TLC5947 LED drivers, and communicates with the Pi 4 over UART.
- **Processor boundary**
  The Pico does not interpret SCPI commands or communicate directly with the oscilloscope. It reports front-panel input events to the Pi 4 and receives LED/status update data from the Pi 4. This keeps all oscilloscope-specific logic on the Pi 4 side while keeping time-sensitive button, encoder, and LED handling local to the PCB.

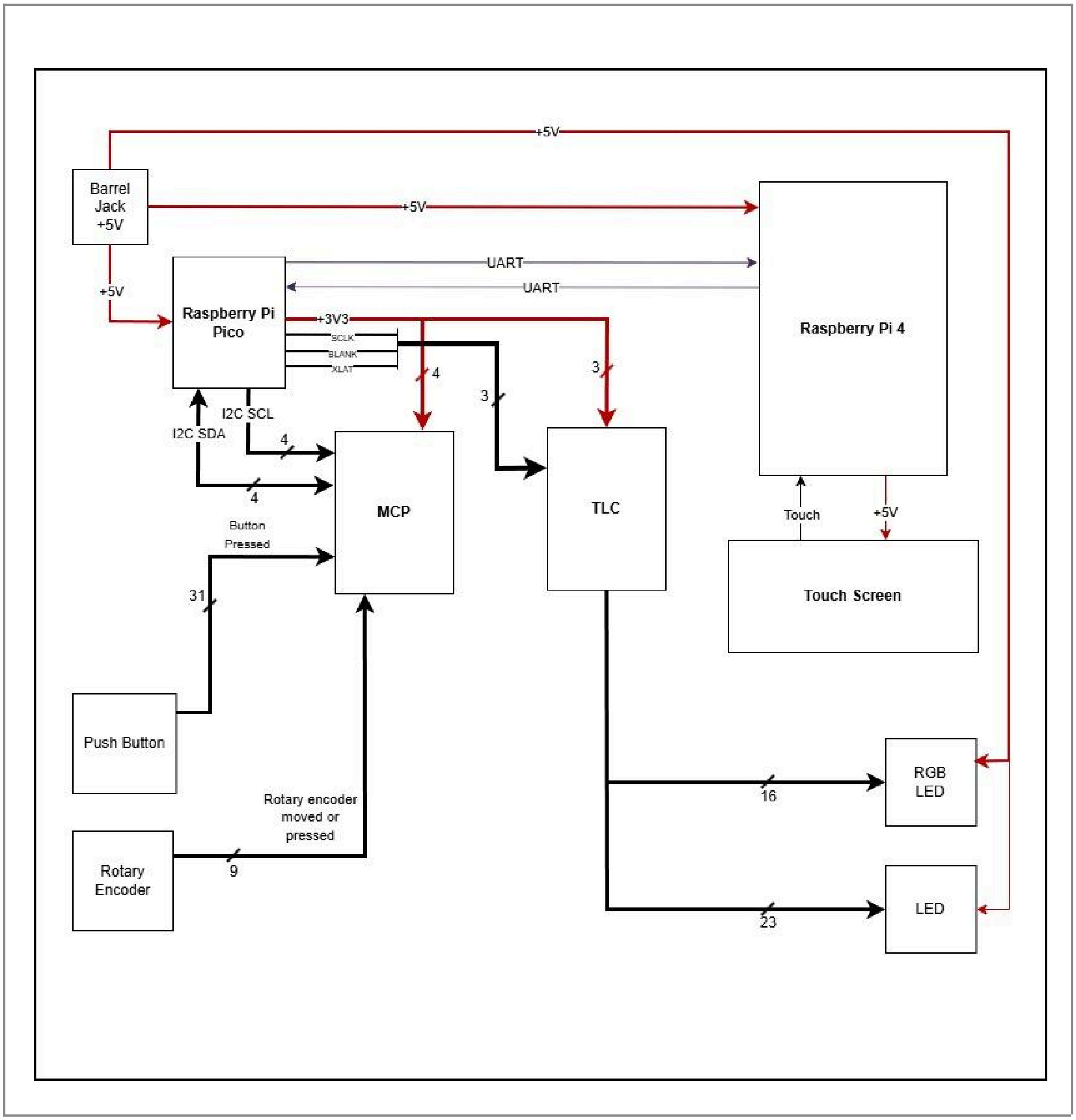


Figure 1.1. Block architecture of the AFP system showing Pi 4, Pico, MCP/TLC subsystems, and external interfaces

1.2 Major Subsystems

Subsystem	Part(s)	Bus / interface	Notes
System controller	Raspberry Pi 4 Model B	UART to Pico; USB/Ethernet external to oscilloscope	Mounted on the back side via 2×20 header + M2.5 standoffs.
Real-time I/O controller	Raspberry Pi Pico / RP2040 module	I²C, SPI, UART	SMD module on the top side
GPIO expansion	4 × MCP23017	I²C addresses 0x20–0x23	Provides input expansion for buttons and encoders
LED drivers	3 × TLC5947	SPI daisy chain (TLC1→TLC2→TLC3)	Provides 72 constant-current LED channels, with 70 used in Rev B
Display connection	4D Systems 4DHDMI-70 (7″, 800×480, cap. touch)	HDMI + USB through Pi 4	Powered from Pi USB rail
Power input	PJ-037AH barrel jack + SLW-1276884 slide switch	5 V / 5 A wall adapter (Qualtek QAWA-30-5)	Powers the AFP board and selected system loads; protection limitations are documented later

1.3 PCB Form Factor

- **Board size:** 214 × 183 mm rectangular PCB, 1.6 mm thick, 4-layer FR-4.
- **Fabrication:** Designed for JLCPCB fabrication with lead-free HASL surface finish.
- **Top side / user-facing side:** All main AFP PCB components are placed on the top side, including push buttons, rotary encoders, LEDs, Raspberry Pi Pico, MCP23017 I/O expanders, TLC5947 LED drivers, passives, test points, and silkscreen labels.
- **Bottom side / rear side:** The Raspberry Pi 4 is mounted on the rear side using standoffs and connected to the AFP PCB using jumper wires/header connections as needed.
- **Mechanical support:** Corner mounting holes were added to support attachment to a rear enclosure, back plate, or stand. This allows the PCB to be mechanically supported without adding a separate top front-panel overlay.
- **PCB-as-front-panel design:** The Rev B PCB serves as the user-facing front panel for this prototype. No separate top overlay or membrane panel is used in this revision; the front-panel labels are provided by the PCB silkscreen.

2. Power Architecture

2.1 Input and Distribution

The Rev B AFP uses a single external 5 V DC input through the PJ-037AH barrel jack (J1, center-positive). The input is routed through the slide switch S2. In the schematic, this switch is currently designated as U45 and should be noted as a designator cleanup item. The switch is used only as the main system ON/OFF switch.

After the switch, the voltage becomes the main switched rail labeled +5V OUT in the schematic. The specified source is the Qualtek QAWA-30-5 5 V / 5 A wall adapter.

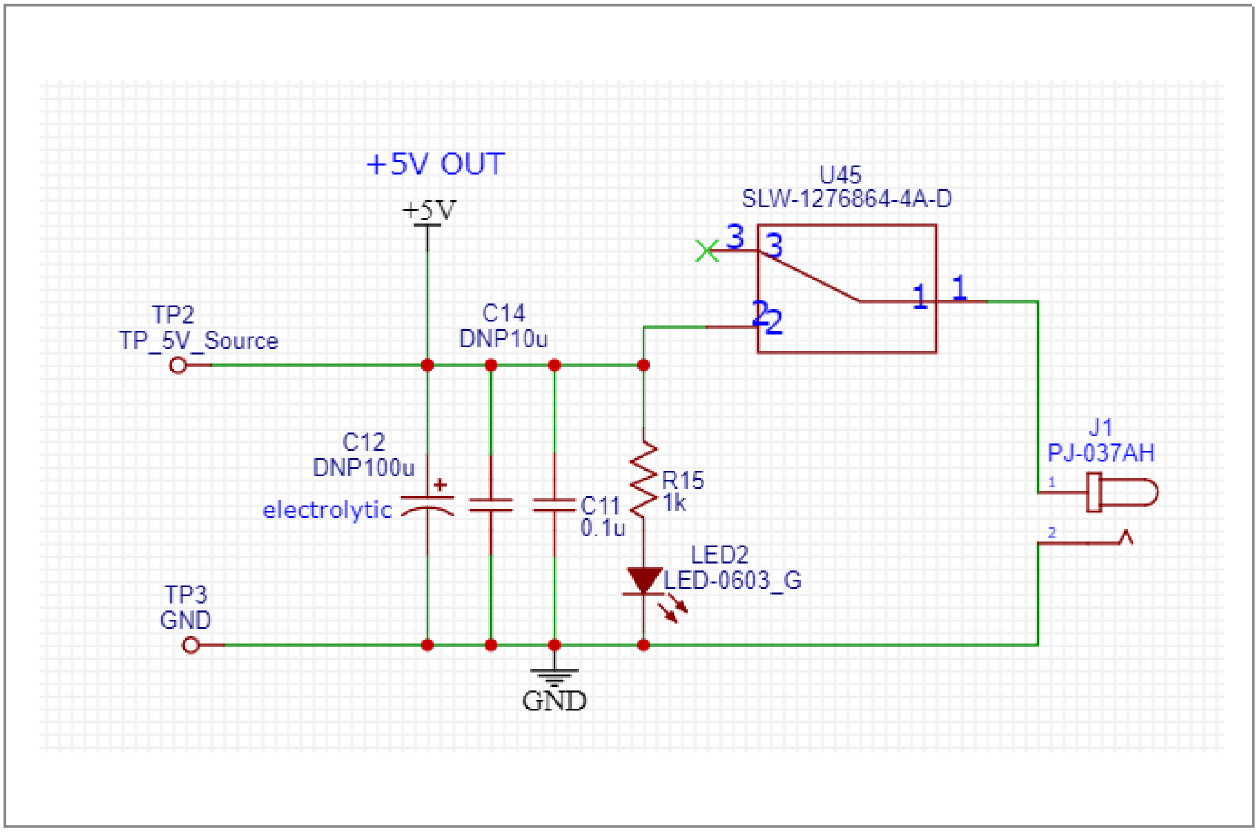


Figure 2.1. Power input circuit — barrel jack, slide switch, decoupling, and protection (schematic sheet 1/9)

From +5V OUT, power is distributed to:

1. Raspberry Pi 4 through R13, a 0 Ω select resistor.
2. Raspberry Pi Pico VSYS through R16, a 0 Ω select resistor.

3. TLC5947 LED anodes, connected directly to the 5 V LED supply rail. LED current is controlled by the TLC5947 constant-current sink outputs, so no individual series resistor is used for each TLC-driven LED channel.
4. Green power indicator LED2 through R15 = 1 kΩ. LED2 turns on when the switch is ON and +5V OUT is active.
5. TP2, which measures +5V OUT after the ON/OFF switch. TP3 is the ground reference test point.

The 3.3 V rail is generated by the Raspberry Pi Pico onboard regulator from VSYS. The Pico 3V3OUT output powers the MCP23017 logic supply, TLC5947 logic supply, and the I²C pull-up rail. No separate 3.3 V LDO is used on the Rev B PCB.

C11 = 0.1 µF is used for local high-frequency decoupling. C12 = 100 µF and C14 = 10 µF are populated to provide additional bulk capacitance and improve power-rail reliability during startup and load changes.

2.2 Power Budget

Load	**Rail**	**Typ (mA)**	**Max (mA)**
Raspberry Pi 4	+5V OUT	1200	3000
7″ HDMI display, powered through Pi USB	+5V via Pi USB	300	500
Raspberry Pi Pico	+5V OUT / VSYS	30	100
4 × MCP23017 (logic)	+3.3V	4	10
3 × TLC5947 (logic VCC only)	+3.3V	6	30
TLC5947 LED current, 72 channels used	+5V out	500	~1050
Green power LED2	+5V out	3	5
Total system	5 V input	~2043 mA / ~2.04 A	~4695 mA / ~4.70 A

The Rev B AFP uses a **5 V / 5 A barrel-jack input** to power the full system. The worst-case budget is close to the supply limit, mainly because the Raspberry Pi 4 and LED system are the largest expected loads. Normal operation is expected to be lower because not all LEDs will be on at full brightness at the same time.

2.3 Decoupling

- Per-IC 100 nF (0603) at every MCP VDD (C5–C8) and every TLC VCC (C1, C3, C4)
- Pico VSYS: C9 = 100 nF, C2 = 100 µF bulk
- Pi 4 +5 V feed: C13 = 100 nF, C10 = 100 µF bulk (PCB-side; Pi 4 has its own internal filtering)
- +5 V input at J1: C11 = 100 nF. C12 = 100 µF electrolytic and C14 = 10 µF are populated to provide additional bulk capacitance and improve 5 V rail stability.

2.4 Power-Source Select (R13, R16)

R13 and R16 are 0 Ω resistors used to connect the main +5V OUT rail to the Raspberry Pi 4 and Raspberry Pi Pico power inputs during normal operation.

1. **R13 — Raspberry Pi 4 power link**
   When R13 is populated, the Raspberry Pi 4 can be powered from the AFP PCB through its 5 V GPIO power connection. In normal system operation, R13 remains populated.
2. **R16 — Raspberry Pi Pico power link**
   When R16 is populated, the Raspberry Pi Pico is powered from the AFP PCB through VSYS. In normal system operation, R16 remains populated.
3. **Normal operating condition**
   For normal AFP operation, the PCB power input is connected, S2 is turned ON, and the +5V OUT rail powers the AFP system.
4. **Pico programming procedure**
   To program the Pico, the AFP PCB power should be unplugged or switched OFF first. The Pico can then be powered and programmed through its USB connection. After programming is complete, unplug the Pico USB connection and return the system to normal PCB-powered operation.
5. **Raspberry Pi 4 programming/setup procedure**
   To work on the Raspberry Pi 4 separately, disconnect the jumper/header power connection from the AFP PCB so the Pi 4 is not being powered from +5V OUT. Then power the Pi 4 externally through its own USB-C power input. After setup or programming is complete, disconnect the external Pi 4 power and reconnect the normal AFP power connection.

Dual-source hazard

Do not power the same board from two 5 V sources at the same time. For example, do not power the Pico from the AFP PCB while also powering it through USB. Also, do not power the Pi 4 from the AFP PCB while also powering it from its own USB-C input. This can create a backfeed path between power sources.

2.5 Protection Gaps

Rev B does not include a fuse, TVS diode, reverse-polarity protection, or ideal-diode power-path protection on the 5 V input. This was accepted for the controlled capstone prototype, but it should be improved in a future revision if the board will be used by more users or outside a supervised lab environment.

Recommended Rev C improvements include:

1. Polyfuse or resettable fuse on the 5 V input.
2. TVS diode for transient protection.
3. Reverse-polarity protection.
4. Ideal-diode or power-path protection to prevent USB/PCB power backfeeding.

2.6 Test Points

TP	Signal	Use
TP1	+5 V at Pi 4 power connection	Confirm Pi 4 is sourced from PCB rail
TP2	+5 OUT after S2	Confirms that the switched 5 V system rail is active
TP3	GND	Reference for all voltage measurements

3. Raspberry Pi Pico Integration

The Raspberry Pi Pico is used as the real-time front-panel I/O controller for the AFP Rev B PCB. It handles local hardware tasks such as reading buttons and encoders, receiving interrupt signals from the MCP23017 I/O expanders, driving the TLC5947 LED driver chain, and communicating with the Raspberry Pi 4.

3.1 Power

VSYS on the Raspberry Pi Pico is fed from the +5V OUT rail through R16, a 0 Ω power link. The Pico onboard regulator generates the local 3.3 V logic rail on 3V3OUT. This 3.3 V rail powers the MCP23017 I/O expanders, TLC5947 logic supply, and I²C pull-up resistors.

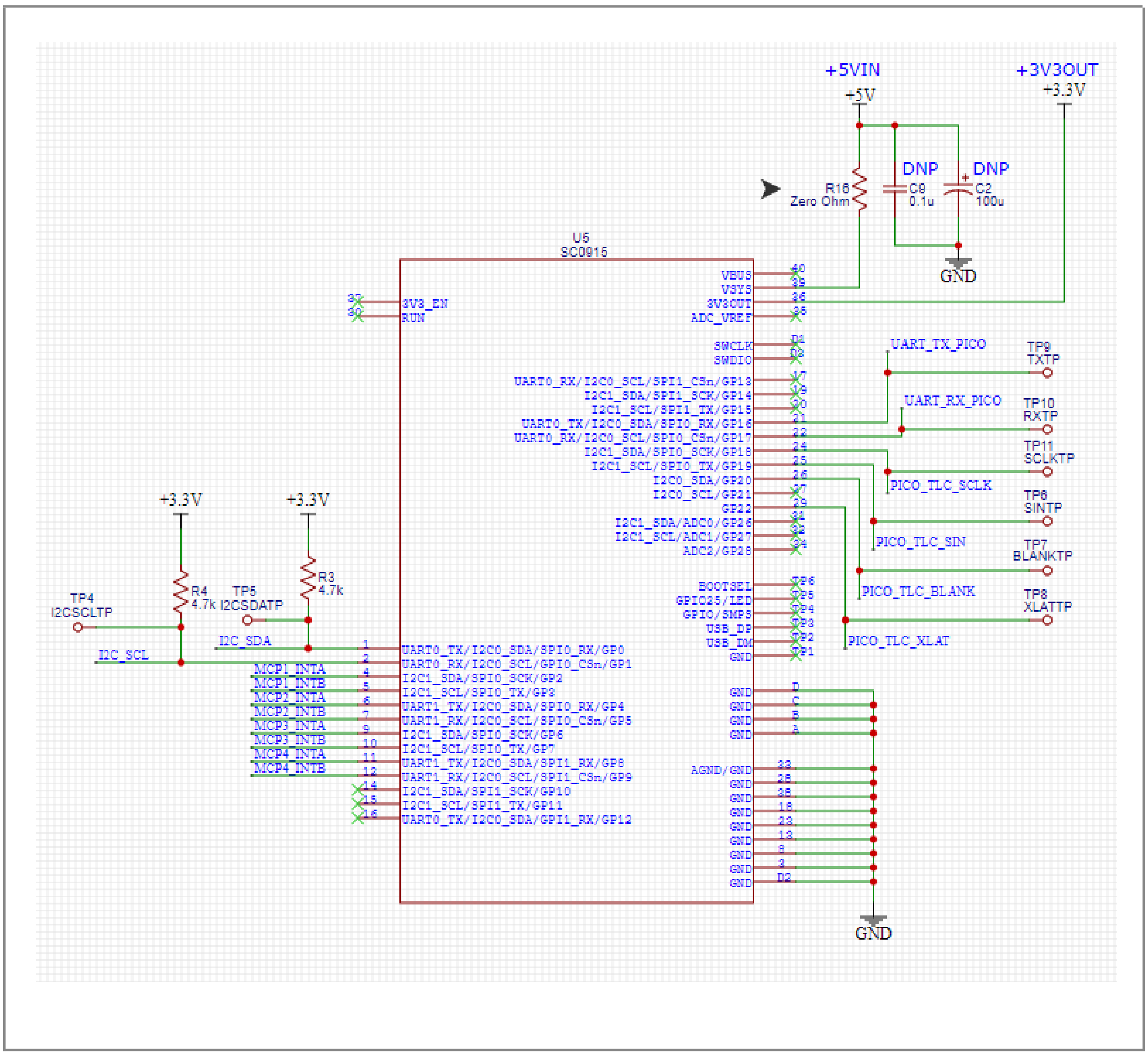


Figure 3.1. Raspberry Pi Pico interface controller and pin assignments (schematic sheet 2/9)

3.2 GPIO Map:

Phys pin	GPIO	Net	Function
1	GP0	I2C_SDA	I²C data line for all four MCP23017 I/O expanders
2	GP1	I2C_SCL	I²C clock line for all four MCP23017 I/O expanders
4	GP2	MCP1_INTA	Interrupt input from MCP1 Port A
5	GP3	MCP1_INTB	Interrupt input from MCP1 Port B
6	GP4	MCP2_INTA	Interrupt input from MCP2 Port A
7	GP5	MCP2_INTB	Interrupt input from MCP2 Port B
9	GP6	MCP3_INTA	Interrupt input from MCP3 Port A
10	GP7	MCP3_INTB	Interrupt input from MCP3 Port B
11	GP8	MCP4_INTA	Interrupt input from MCP4 Port A
12	GP9	MCP4_INTB	Interrupt input from MCP4 Port B
21	GP16	UART_TX_PICO	UART transmit signal from Pico to Raspberry Pi 4
22	GP17	UART_RX_PICO	UART receive signal from Raspberry Pi 4 to Pico
24	GP18	PICO_TLC_SCLK	Serial clock signal for the TLC5947 LED driver chain
25	GP19	PICO_TLC_SIN	Serial data input to the first TLC5947 LED driver
26	GP20	PICO_TLC_BLANK	Output enable/blanking control for the TLC5947 drivers
29	GP22	PICO_TLC_XLAT	Latch signal for updating TLC5947 LED output data
36	3V3OUT	+3.3V	Powers MCPs, TLC logic, pull-ups
39	VSYS	+5V (via R16)	Pico power input

3.3 Test Points

TP	Signal	TP	Signal
TP4	I2C_SCL	TP9	UART_TX_PICO (Pico→Pi)
TP5	I2C_SDA	TP10	UART_RX_PICO (Pi→Pico)
TP6	PICO_TLC_SIN	TP11	PICO_TLC_SCLK
TP7	PICO_TLC_BLANK		
TP8	PICO_TLC_XLAT		

3.5 UART to Raspberry Pi 4

The Raspberry Pi Pico communicates with the Raspberry Pi 4 through a direct 3.3 V UART connection. No level shifter is required because both devices use 3.3 V GPIO logic.

1. Pico transmit line:
   UART_TX_PICO is driven by the Pico and connects to the Raspberry Pi 4 UART receive pin.
2. Pico receive line:
   UART_RX_PICO is driven by the Raspberry Pi 4 UART transmit pin and connects to the Pico receive pin.
3. Cross-wiring:
   The UART lines are cross-connected as required for serial communication:
   - Pico TX → Pi 4 RX
   - Pi 4 TX → Pico RX
4. Common ground:
   The Pico and Raspberry Pi 4 share the same ground reference through the AFP PCB power and ground connections.
5. Hardware note:
   The UART nets are named from the Pico side. Therefore, UART_TX_PICO means the signal transmitted by the Pico, and UART_RX_PICO means the signal received by the Pico.
6. Firmware note:
   Baud rate, frame format, and protocol behavior are defined in firmware and are not specified in this hardware reference section.

4. Raspberry Pi 4 Integration

The Raspberry Pi 4 connects to the AFP PCB through the GPIO1 2×20 header. In Rev B, only the power, ground, and UART pins are used. All other GPIO pins are not connected to AFP circuitry.

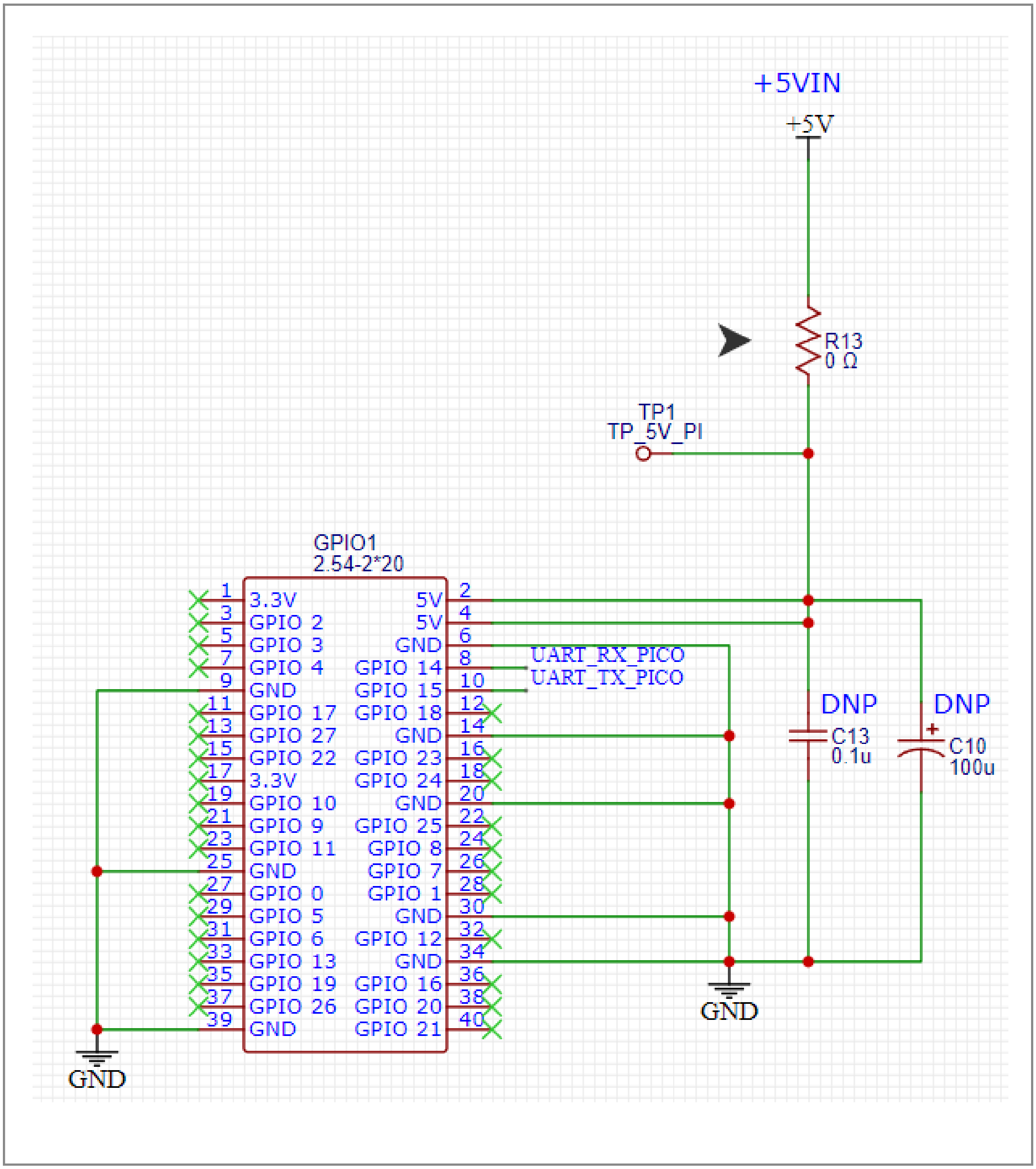


Figure 4.1. Raspberry Pi 4 controller and display interface (schematic sheet 3/9)

4.1 Header Pin Usage

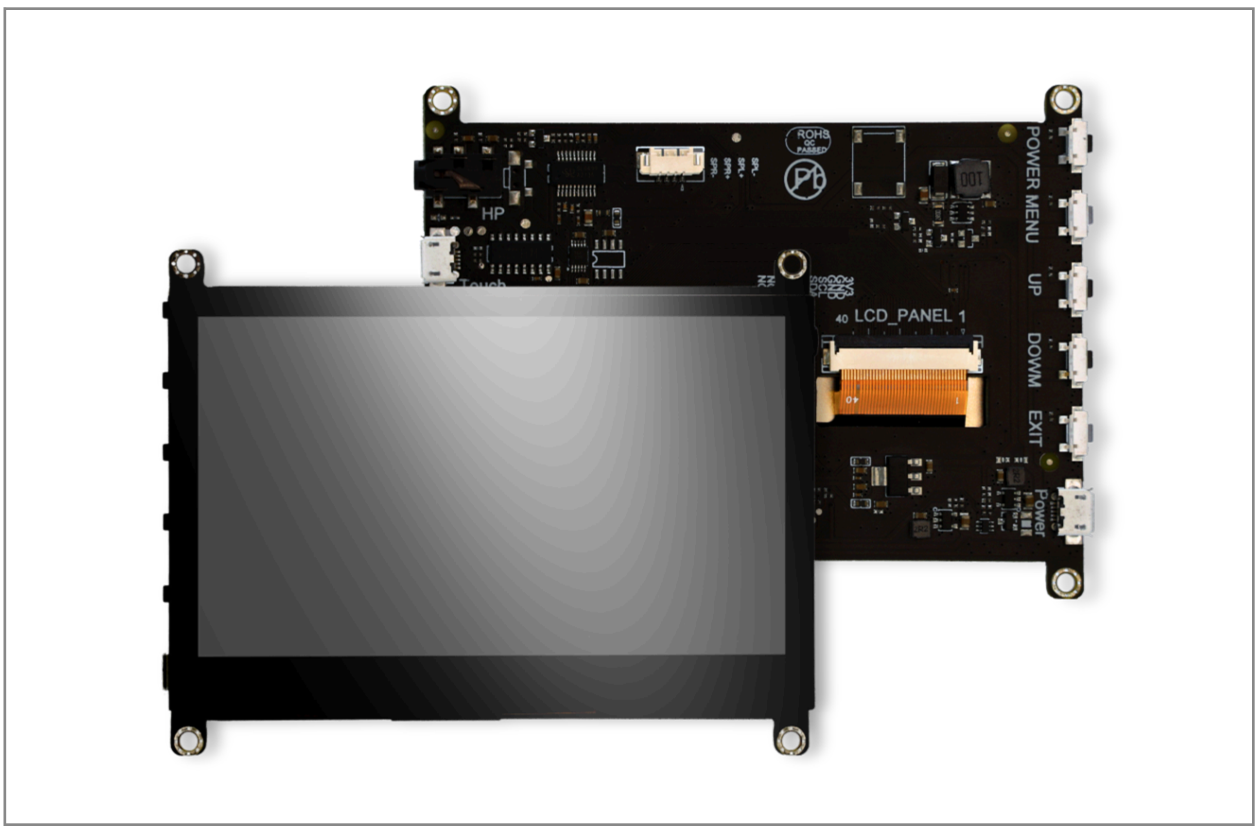
Pi pin	Pi function	AFP net	Notes
2, 4	+5V input	+5V OUT through R13	Powers the Raspberry Pi 4 from the AFP PCB during normal operation
6, 14, 20, 25, 30, 34, 39	GND	GND	Common ground
8	GPIO14 / TXD	UART_RX_PICO	Pi 4 transmit line to Pico receive
10	GPIO15 / RXD	UART_TX_PICO	Pi 4 receive line from Pico transmit

All other Pi GPIO pins are unconnected on the AFP. USB, HDMI, Ethernet, SD card used directly — not routed through the PCB.

4.2 Display Connection

4D Systems 4DHDMI-70 (7″, 800×480, capacitive touch, 4.8–5.2 V) connects to the Pi 4 externally:

- Video: HDMI cable from Raspberry Pi 4 to the display HDMI input.
- Touch interface: USB cable from Raspberry Pi 4 to the display USB port.
- Display power: The display receives power through the USB connection from the Raspberry Pi 4.



5. MCP23017 I/O Expansion

Four MCP23017T-E/SO devices on a shared I²C bus. Each provides 16 GPIO (two 8-bit ports A and B). Total 64 input lines; ~56 used in Rev B.

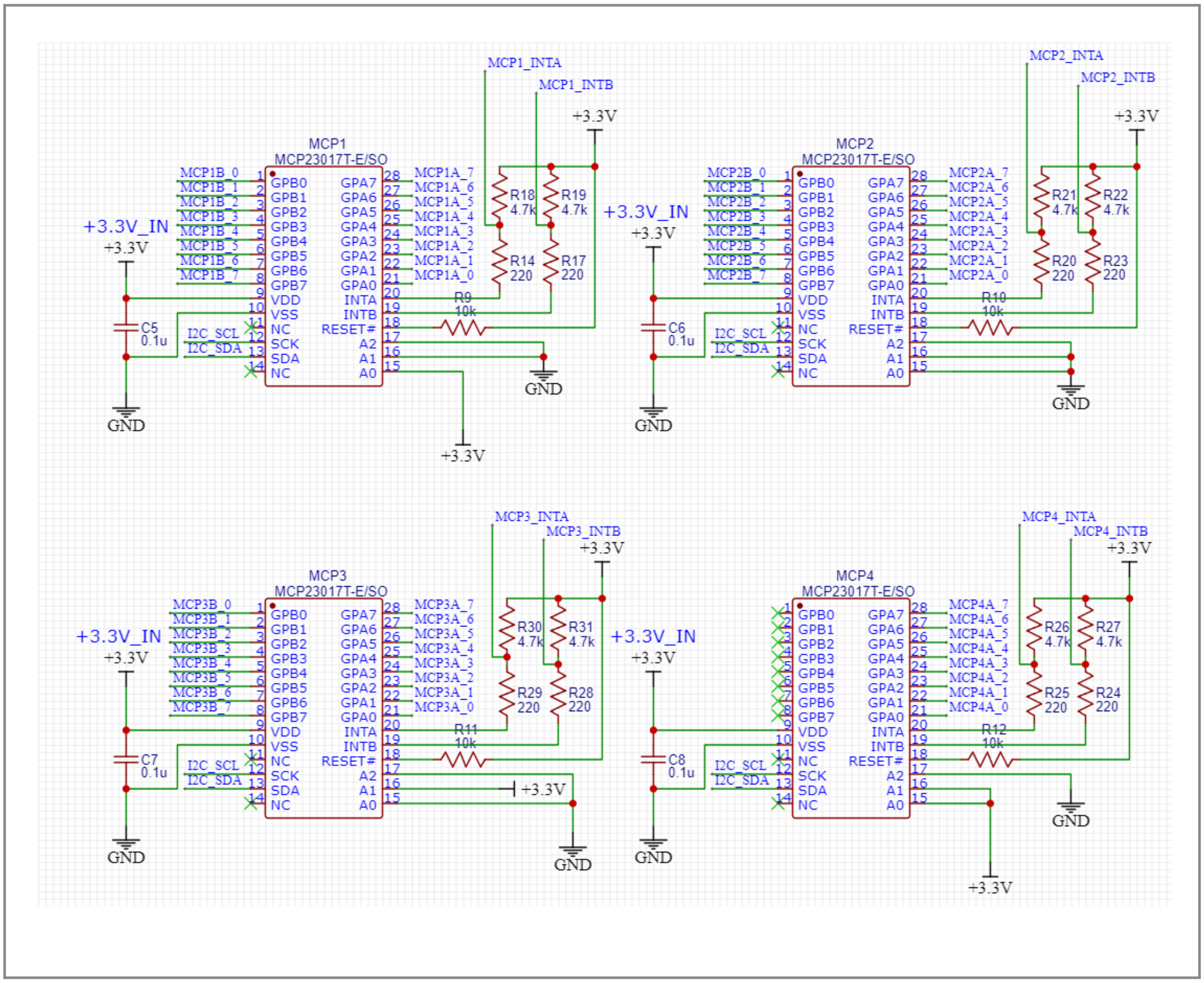


Figure 5.1. Four MCP23017 I²C GPIO expanders with addressing, pull-ups, and interrupt routing (schematic sheet 4/9)

5.1 Bus and Addressing

MCP23017 I/O expanders are powered from the Pico 3.3 V rail and connected to the shared I²C bus using I2C_SCL and I2C_SDA. Each MCP23017 provides 16 programmable GPIO pins, arranged as two 8-bit ports: Port A and Port B. In this design, these GPIO pins are used for reading front-panel push buttons and rotary encoders.

The I²C bus is pulled up to 3.3 V through R= 4.7 kΩ on SCL and R = 4.7 kΩ on SDA.

Each MCP23017 uses a unique hardware address through pins A0, A1, and A**2**.

Device	Address	Section (per schematic)
MCP1	**0x21**	**Acquisition/Trigger buttons + Cursor A/B encoders + start of Trigger Level**
MCP2	**0x20**	**Rest of Trigger Level encoder + Vertical Position/Scale encoders + 8 channel buttons**
MCP3	**0x22**	**Math/Ref/Bus + all Horizontal encoders + Zoom/Pan + navigation**
MCP4	**0x23**	**4 macros + 4 system buttons (Touch Off / User / Default Setup / Autoset)**
Hardware note: MCP1 and MCP2 labels are not in sequential address order in Rev B. The schematic labels should be followed as the reference, and firmware should use the addresses shown in this table.		

5.2 Interrupt Routing

Each MCP's INTA (Port A) and INTB (Port B) routed to dedicated Pico GPIOs. Eight lines on a contiguous block (GP2–GP9). Enables event-driven reads instead of polling.

5.3 Support Components (per device)

- Decoupling: One 0.1 µF capacitor is placed across VDD and VSS for each MCP23017.
- Reset pull-up: RESET is pulled up to +3.3 V through a 10 kΩ resistor.
- I²C pull-ups: The shared I²C bus uses 4.7 kΩ pull-up resistors to +3.3 V on SDA and SCL.
- Interrupt pull-ups: Each INTA and INTB line has a 4.7 kΩ pull-up resistor to +3.3 V.
- Interrupt series resistors: Each INTA and INTB line includes a 220 Ω series resistor before the Pico GPIO input. These resistors provide small current limiting and help protect the GPIO pins during transitions or accidental contention.
- Address pins: tied to GND or +3.3 V per table above.

5.4 Allocation Summary

Full pin-by-pin allocation in Appendix A. Summary by function:

- All buttons: active-LOW, switch closes to GND. Firmware MUST enable internal pull-ups (GPPUx = 1) on every used pin.
- Encoder rotation signals: Rotary encoder phase A and phase B signals are connected to MCP23017 input pins and are read as active-LOW digital inputs.
- Encoder push switches: Encoders with push switches use a separate MCP23017 input for the push function. The push switch also closes to GND when pressed.
- Trigger Level encoder: The Trigger Level encoder is split across two MCP23017 devices. Its phase A signal is on MCP1, while the remaining phase/switch signals are on MCP2. This should be handled carefully in firmware and verified against Appendix A.
- Zoom and Pan encoders: Zoom and Pan use encoder rotation signals only. Their push-switch function is not wired through the encoder body. The Zoom push function is handled by a separate Zoom button.
- Unused inputs: Unused MCP23017 pins should remain configured safely in firmware. MCP4 Port B is unused in Rev B and may be available for future expansion, if confirmed by the final schematic.

5.5 Firmware Init Requirements

The MCP23017 devices must be initialized in firmware before the front-panel inputs and interrupt lines operate correctly:

- Direction: Configure all used MCP23017 GPIO pins as inputs.
- Input pull-ups: Enable internal pull-ups on all used button and encoder input pins. Without pull-ups, unused or open switch states may float and cause false triggers.
- Interrupt enable: Enable interrupt-on-change for the used input pins if using the Rev B interrupt routing.
- Interrupt outputs: Configure INTA and INTB as separate interrupt outputs to match the Rev B hardware wiring.
- Interrupt polarity/type: Configure the interrupt output behavior to match the firmware expectation. The Rev B hardware includes external 4.7 kΩ pull-ups on the interrupt lines and 220 Ω series resistors before the Pico GPIO inputs.
- Interrupt clearing: After an interrupt occurs, firmware must read the appropriate MCP23017 GPIO or interrupt capture register to clear the interrupt condition.

6. TLC5947 LED Driver System

The Rev B AFP uses three TLC5947 LED driver ICs to control the front-panel LED indicators. Each TLC5947 provides 24 constant-current sink outputs, giving a total of 72 LED channels.

The TLC5947 devices are used for LED control only. They do not read buttons or communicate with the oscilloscope.

- Main strength: Large LED count is handled with only a few Pico control lines instead of using many GPIO pins.
- LED current control: The TLC5947 provides constant-current sinking, so the LED brightness is controlled by the driver instead of using individual series resistors for every LED.
- Power split: LED anodes are powered from +5V OUT, while TLC5947 logic is powered from the Pico 3.3 V rail.
- Daisy-chain design: The three TLC5947 drivers are connected in series, allowing one Pico serial interface to update all LED channels.
- Hardware role: This design keeps the LED system organized, expandable, and easier to route compared with driving each LED directly from the Pico or MCP23017 expanders.

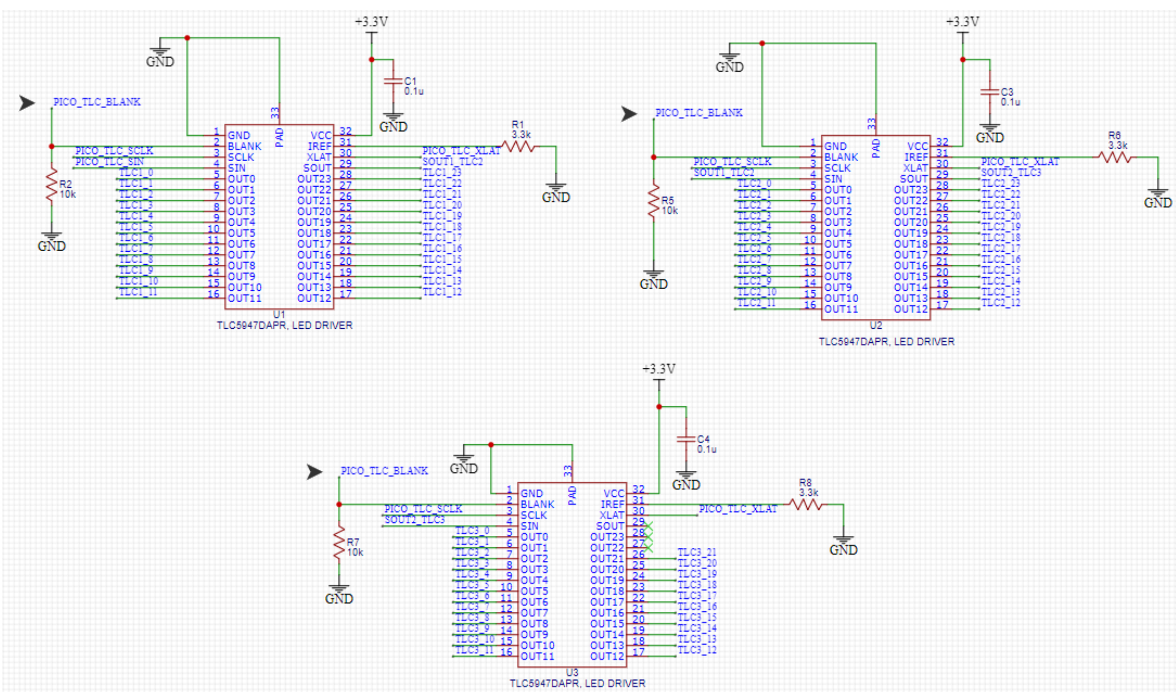


Figure 6.1. TLC5947 LED driver chain — three devices, shared SCLK/XLAT/BLANK, daisy-chained SIN/SOUT (schematic sheet 5/9)

6.1 Chain Topology

Signal	Pico GPIO	Routing
SIN (data)	GP19 (SPI0 TX)	Pico serial data output to TLC1 SIN. TLC1 SOUT connects to TLC2 SIN, and TLC2 SOUT connects to TLC3 SIN.
SCLK (clock)	GP18 (SPI0 SCK)	Shared serial clock from the Pico to all three TLC5947 drivers.
XLAT (latch)	GP22	Shared latch signal from the Pico to all three TLC5947 drivers.
BLANK (output disable)	GP20	Shared output blanking/control signal from the Pico to all three TLC5947 drivers.

The three TLC5947 devices are daisy-chained so the Pico can update all LED channels using one serial data path, one clock line, one latch line, and one blanking control line.

6.2 Output Current

Each TLC5947 sets its LED sink current using an external RIREF resistor. In Rev B, R1, R6, and R8 are 3.3 kΩ, giving an estimated output current of about 12 mA per channel. The LED anodes are connected to +5V OUT, while the TLC5947 outputs sink current through the LED cathodes. With 70 used channels, the worst-case LED current is approximately: 70 channels × 12 mA ≈ 840 mA This value is used as an estimated worst-case LED current for the power budget.

> **Bring-Up/Test**
> The datasheet has two different current formulas depending on which reference. Confirm actual I_CH with a current probe on one channel of each TLC at full PWM (4095/4095). Adjust RIREF if needed.

6.3 BLANK Behavior

- Each TLC has its own 10 kΩ BLANK pull-up to +3.3 V (R2 on TLC1, R5 on TLC2, R7 on TLC3). Confirmed in netlist — not a shared pull-up.
- Before Pico boots: BLANK held high by pull-ups, all outputs disabled, LEDs dark.
- Normal operation: Pico drives BLANK low after the first valid frame is shifted in and latched.

6.4 LED Allocation Summary

70 channels populated, 2 channels (TLC3 OUT22, OUT23) unused and available for future indicators. Full TLC-channel-to-LED mapping in Appendix B.

Device	Used channels	Function summary
TLC1 (U1)	24 / 24	Trigger Level RGB ring, Run/Stop RGB, acquisition indicators, cursor encoder rings, slope/mode/trig'd indicators, Math/Ref/Bus button LEDs
TLC2 (U2)	24 / 24	Vertical Position RGB, Vertical Scale RGB, channel 1–6 RGB LEDs
TLC3 (U3)	22 / 24	Channel 7–8 RGB, Zoom indicator, macro 1–4 RGB, Scope Connected, Horizontal Scale fine, Touch Off. CH22 and CH23 unused.

RGB LEDs are common-anode 4-pin packages. Anode (pin 2) ties to +5 V. Color cathodes: pin 4 = Blue, pin 3 = Green, pin 1 = Red. In the TLC channel sequence for each RGB LED, the lowest channel drives B, middle G, highest R.

6.5 Decoupling and Thermal

Each TLC5947 has a local 100 nF decoupling capacitor placed at its VCC pin. The TLC5947 thermal pad is tied to GND and connected to the ground plane using 9 thermal vias per device. This improves heat transfer from the IC package into the PCB copper during LED operation.

The estimated LED current is about 12 mA per channel, so the thermal vias provide important support when many LED channels are active at high brightness. For Rev B, the thermal pad and via connection are included as part of the LED driver thermal management design.

Per-section schematic sheets

The control schematics are organized by the oscilloscope section. The figures below correspond to the schematic sheets 6/9 through 9/9 of the EasyEDA project.

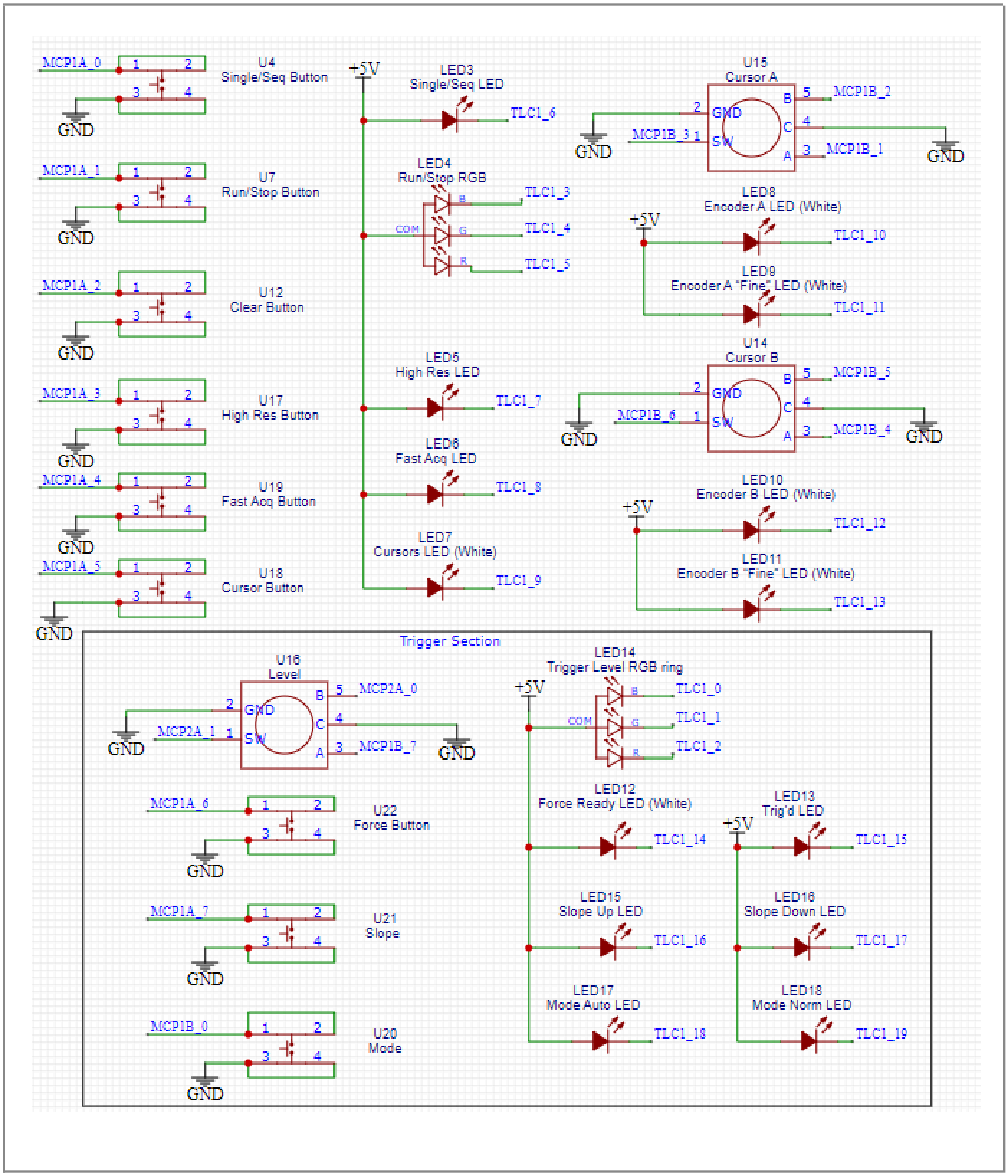


Figure 6.2. Acquisition and trigger controls — Run/Stop, Single/Seq, Clear, Cursors A/B, Trigger Level encoder, Slope/Mode/Force buttons, and associated LEDs (schematic sheet 6/9)

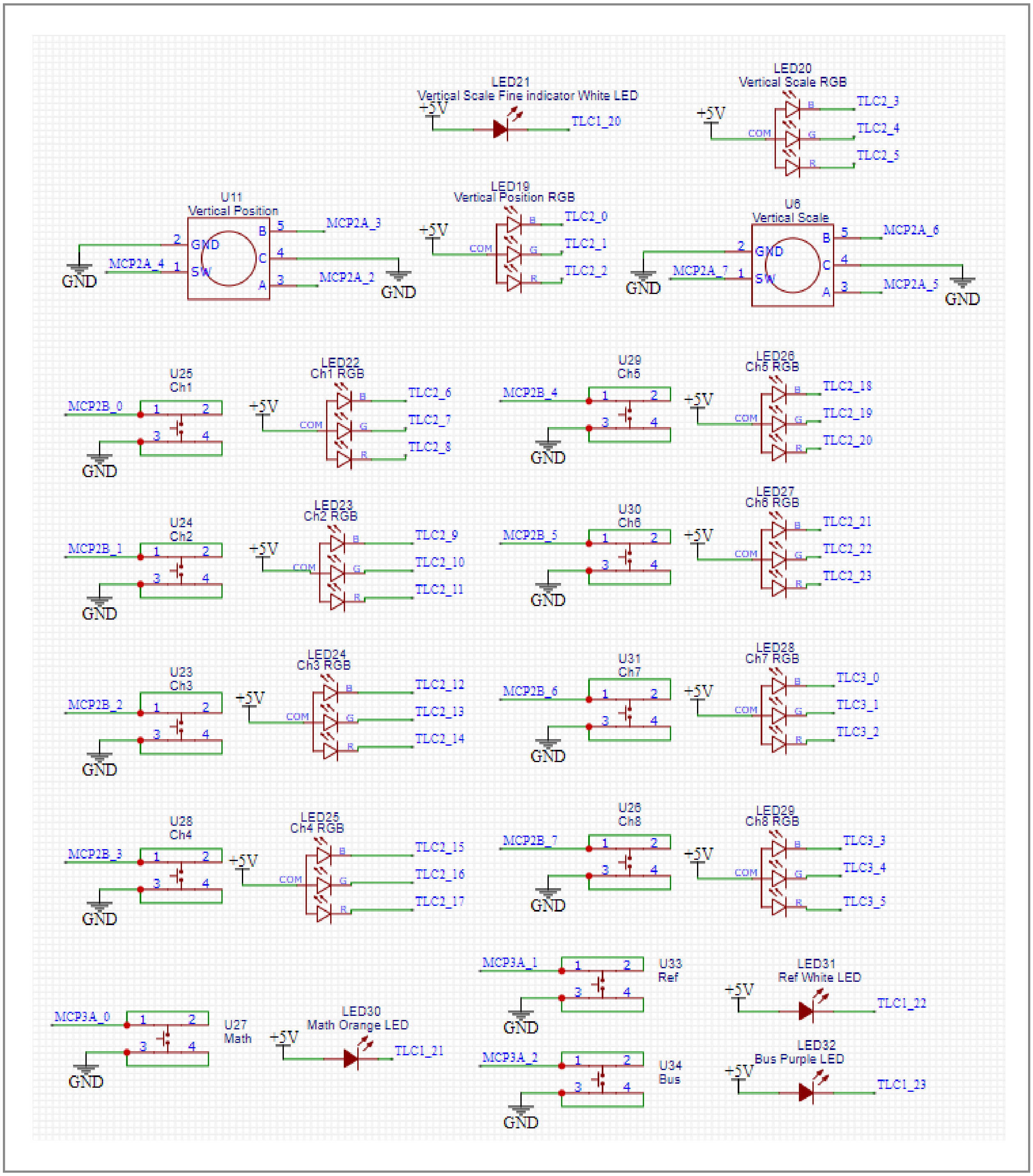


Figure 6.3. Vertical controls — Vertical Position/Scale encoders, channel 1–8 buttons with RGB LEDs, Math/Ref/Bus buttons (schematic sheet 7/9)

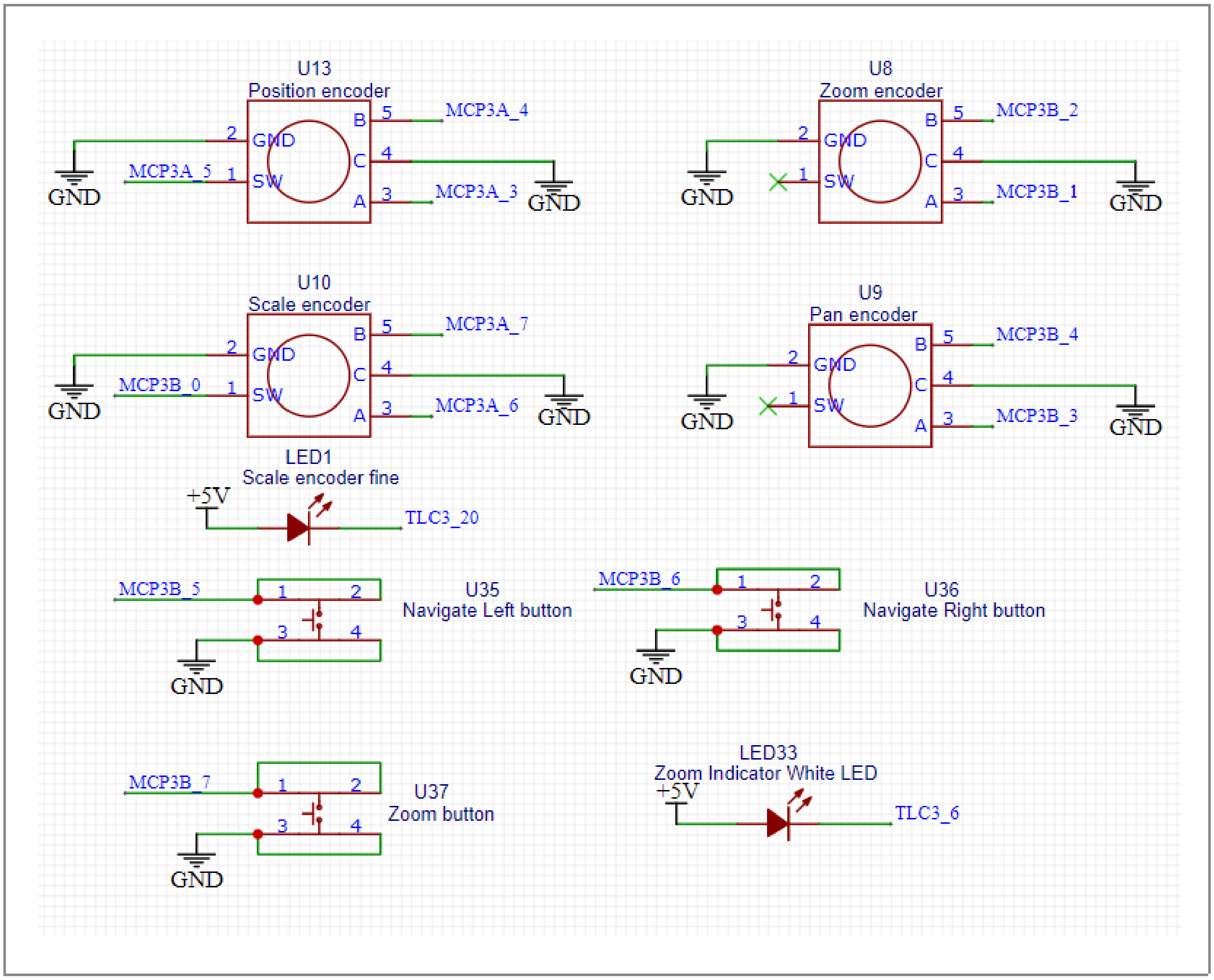


Figure 6.4. Horizontal controls — Horizontal Position/Scale encoders, Zoom and Pan encoders, Navigate Left/Right buttons (schematic sheet 8/9)

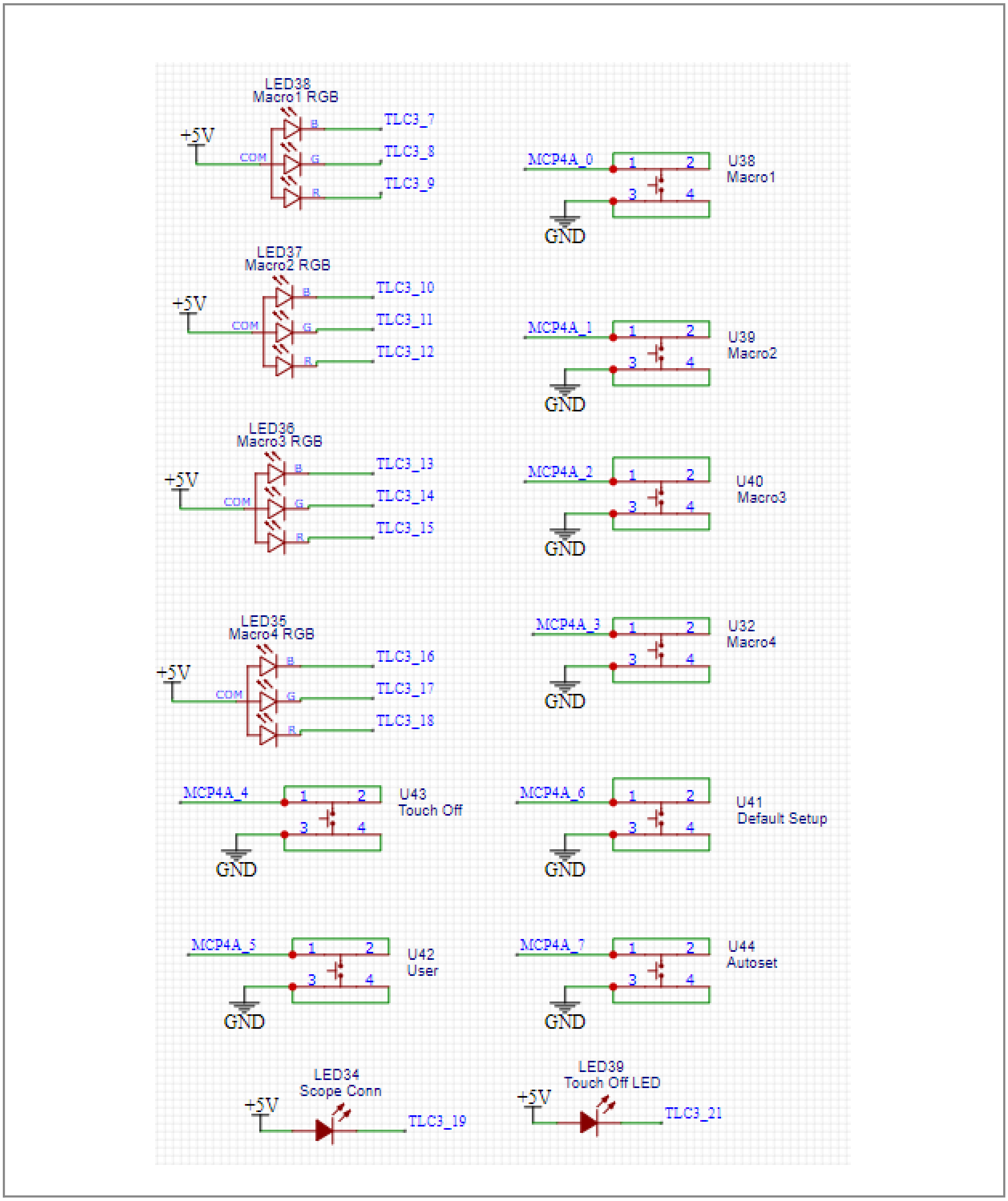


Figure 6.5. System and macro controls — four programmable macros, Touch Off, User, Default Setup, Autoset (schematic sheet 9/9)

7. Control Interface (Buttons & Encoders)

7.1 Inventory

Category	Count	Designators
Encoders with push switch	7	U6, U10, U11, U13, U14, U15, U16
Encoders rotation only	2	U8 (Zoom), U9 (Pan)
Channel buttons	8	U23–U26, U28–U31
Acquisition/Trigger buttons	9	U4, U7, U12, U17–U22
Math/Ref/Bus	3	U27, U33, U34
Horizontal section buttons	3	U35, U36, U37
Macro buttons	4	U32, U38, U39, U40
System buttons	4	U41, U42, U43, U44

Total: Rev B includes 31 standalone push buttons, 9 rotary encoders, and 7 encoder push switches, giving 47 user-facing control actions.

7.2 Electrical Convention

All push buttons are wired as **active-LOW** inputs. One side of each button is connected to **GND**, and the other side is connected to an MCP23017 input. The MCP23017 internal pull-up resistors must be enabled in firmware so the input reads HIGH when open and LOW when pressed.

Each 5-pin rotary encoder uses the same active-LOW input convention:

Encoder pin	Function	Wired to
1	Push switch signal	MCP23017 input
2	Common	GND
3	Phase A	MCP23017 input
4	Common	GND
5	Phase B	MCP23017 input

No hardware debounce circuit is included in Rev B. Button debounce and rotary encoder decoding are handled in firmware.

7.3 Mechanical

All front-panel buttons, rotary encoders, LEDs, and labels are placed on the PCB top side, which acts as the user-facing control surface.

- Buttons and encoders are soldered directly to the PCB.
- Encoder retaining tabs provide additional mechanical support.
- Silkscreen labels are placed near the controls for identification.
- Rev B does not use a separate top overlay or membrane panel.
- Mounting holes are included for PCB support, rear enclosure/back plate attachment, display support, and stand mounting as needed.

7.4 ESD Note

Rev B does not include dedicated board-level ESD protection on the user-operated push-button and encoder input lines. The MCP23017 devices include internal device-level protection, which is acceptable for a controlled capstone prototype. For a future product revision, TVS diode arrays may be added to improve robustness against static discharge during repeated user operation.

8. PCB Design

8.1 Spec Summary

Parameter	Value
Outline	214.25 × 183.26 mm PCB with rounded corners
Layers	4 (Signal / GND / PWR / Signal)
Thickness	1.6 mm FR-4
Surface finish	HASL with lead
Vendor	JLCPCB
Min trace/space	6 mil / 6 mil (JLC standard)
Min drill	0.3 mm

The PCB was ordered as a 4-layer FR-4 board from JLCPCB. The final uploaded Gerber files were detected by JLCPCB as a 183.26 × 214.25 mm board. A blue solder mask with white silkscreen was selected because the PCB is also used as the visible front panel of the prototype, so the color contrast helps make the labels and control sections easier to read.

8.2 Stackup

Layer	Role
L1 Top	Signal + user-facing controls + silkscreen labels
L2 Inner	GND plane (continuous pour)
L3 Inner	PWR plane (+5 V and +3.3 V)
L4 Bottom	Signal

The board uses a Signal/GND/PWR/Signal stackup. This was selected because the top layer carries many front-panel controls and digital signal runs, including the I^2C bus to the MCP23017 expanders and the TLC5947 LED-driver control lines. Placing a ground plane directly below the top signal layer gives these signals a cleaner return path and helps reduce routing noise. The inner power layer is used to distribute the +5 V and +3.3 V rails across the board.

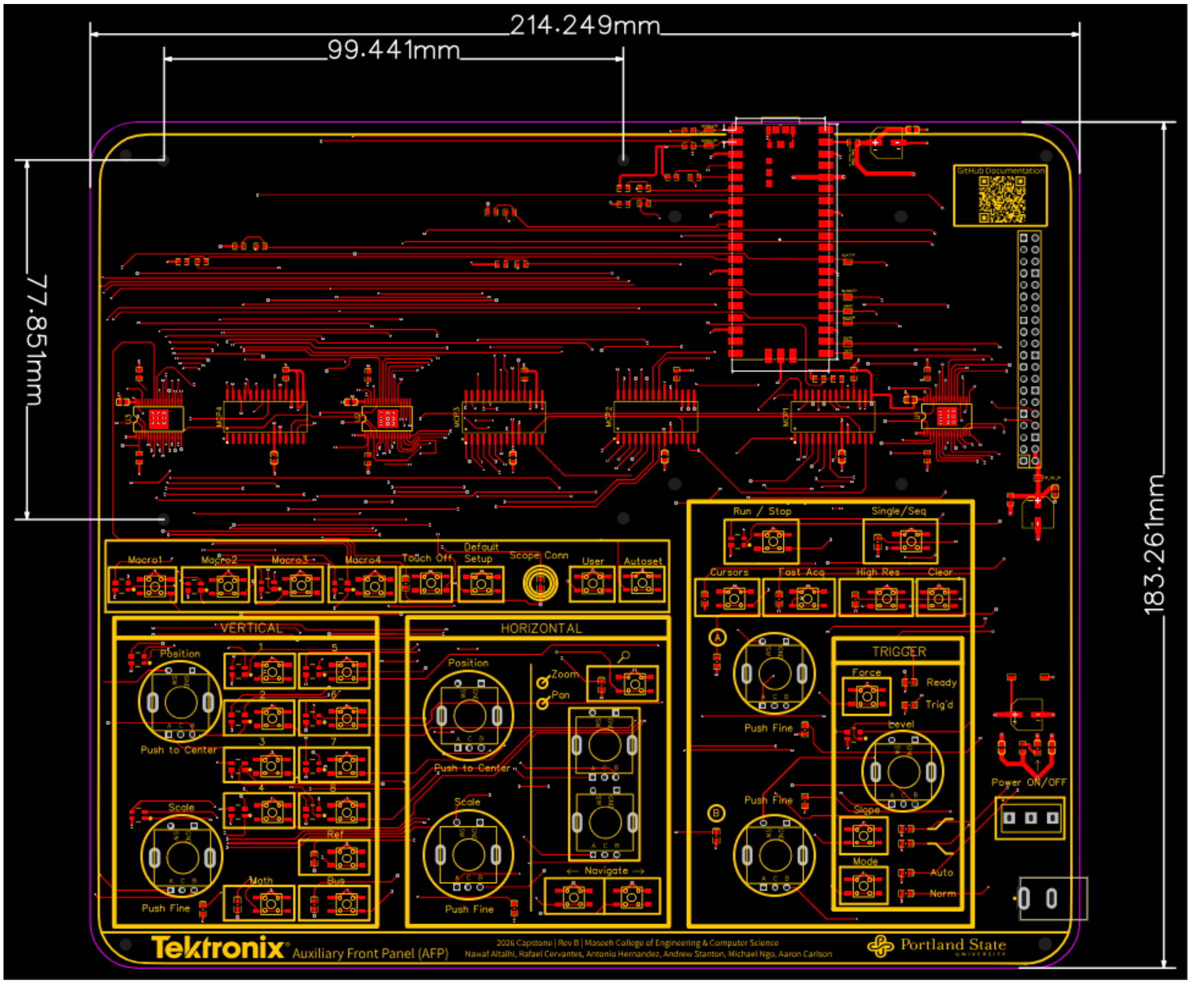


Figure 8.1. L1 Top layer — routing and component placement

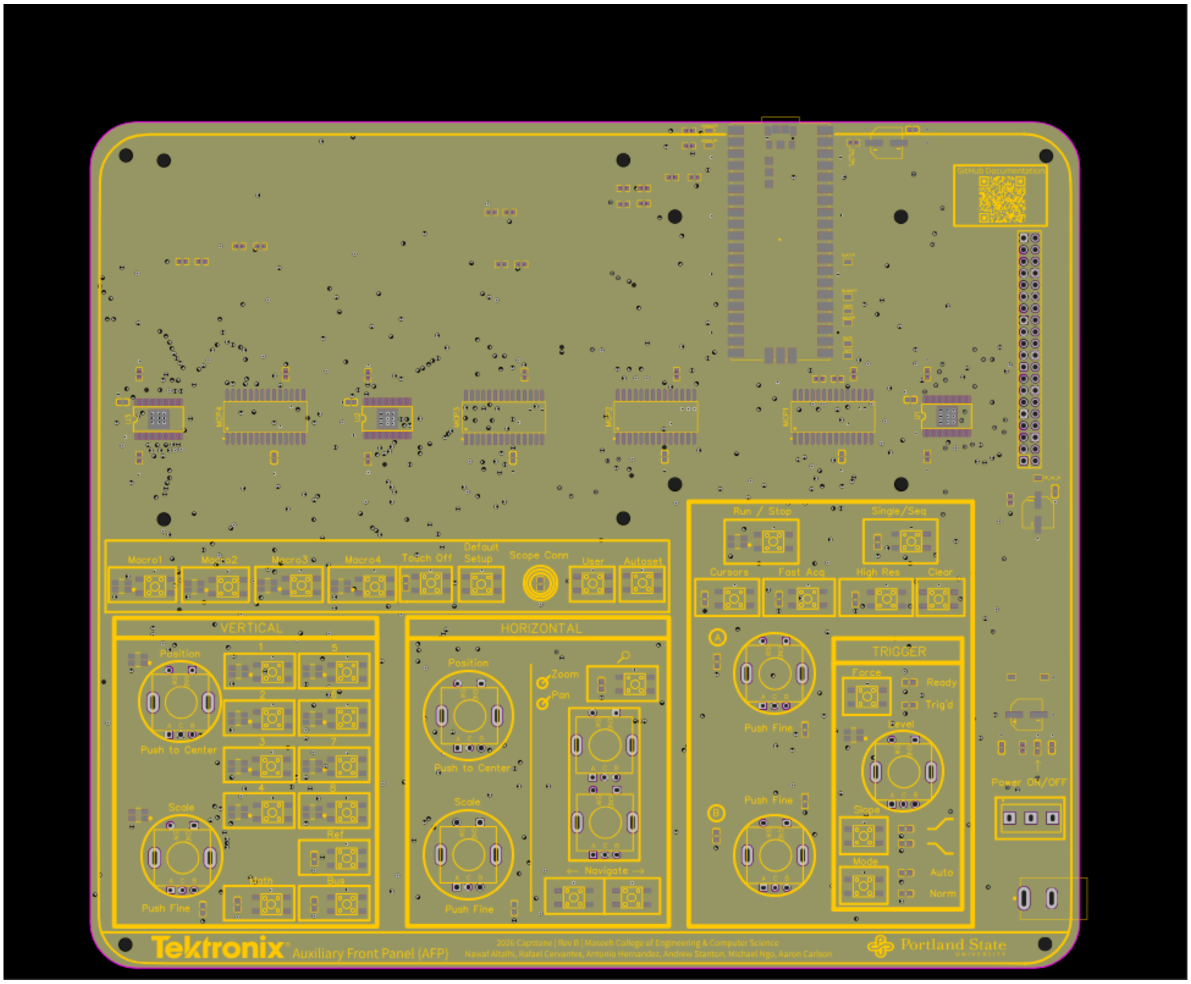


Figure 8.2. L2 Inner — GND plane

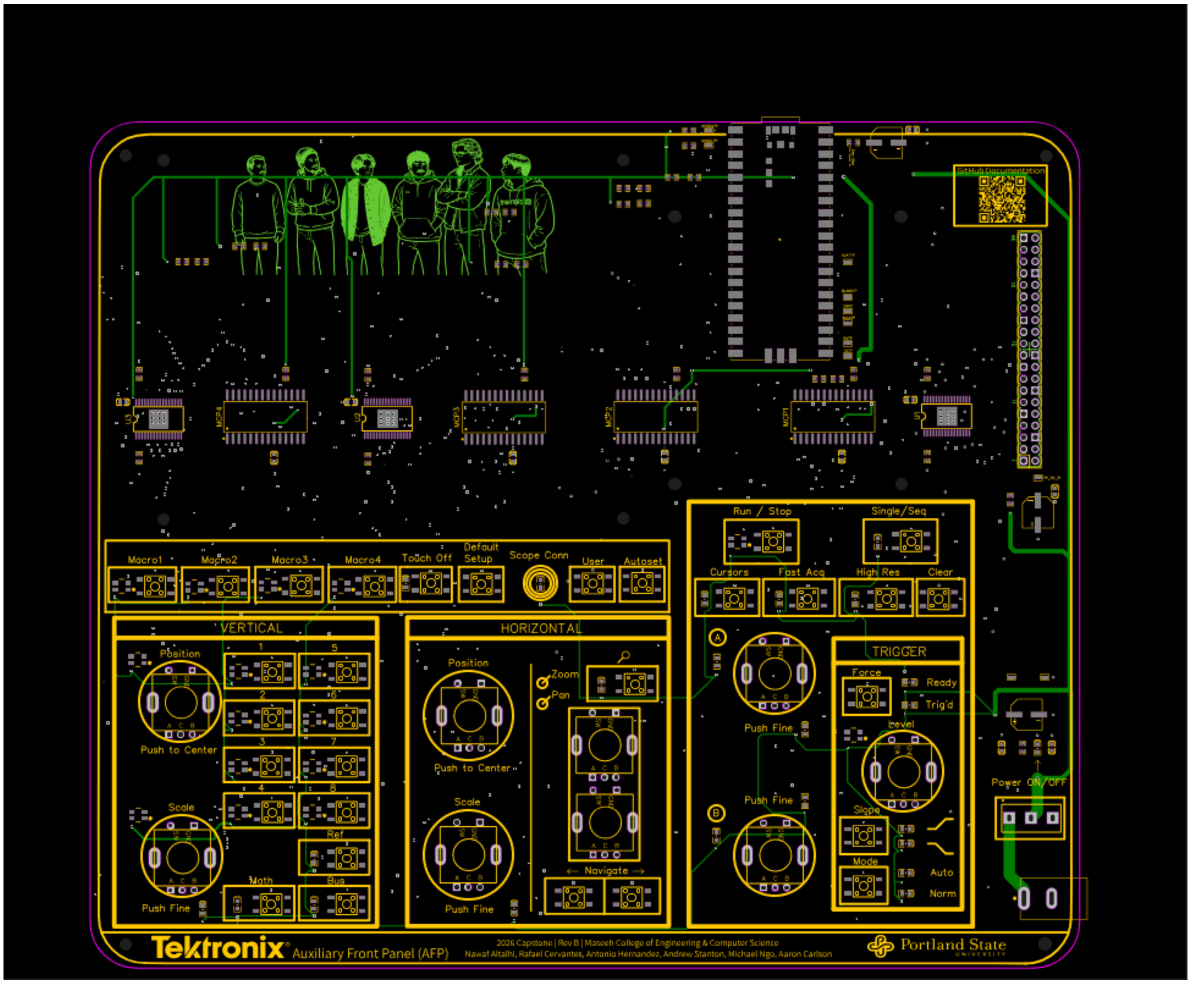


Figure 8.3. L3 Inner — PWR plane with +5 V and +3.3 V

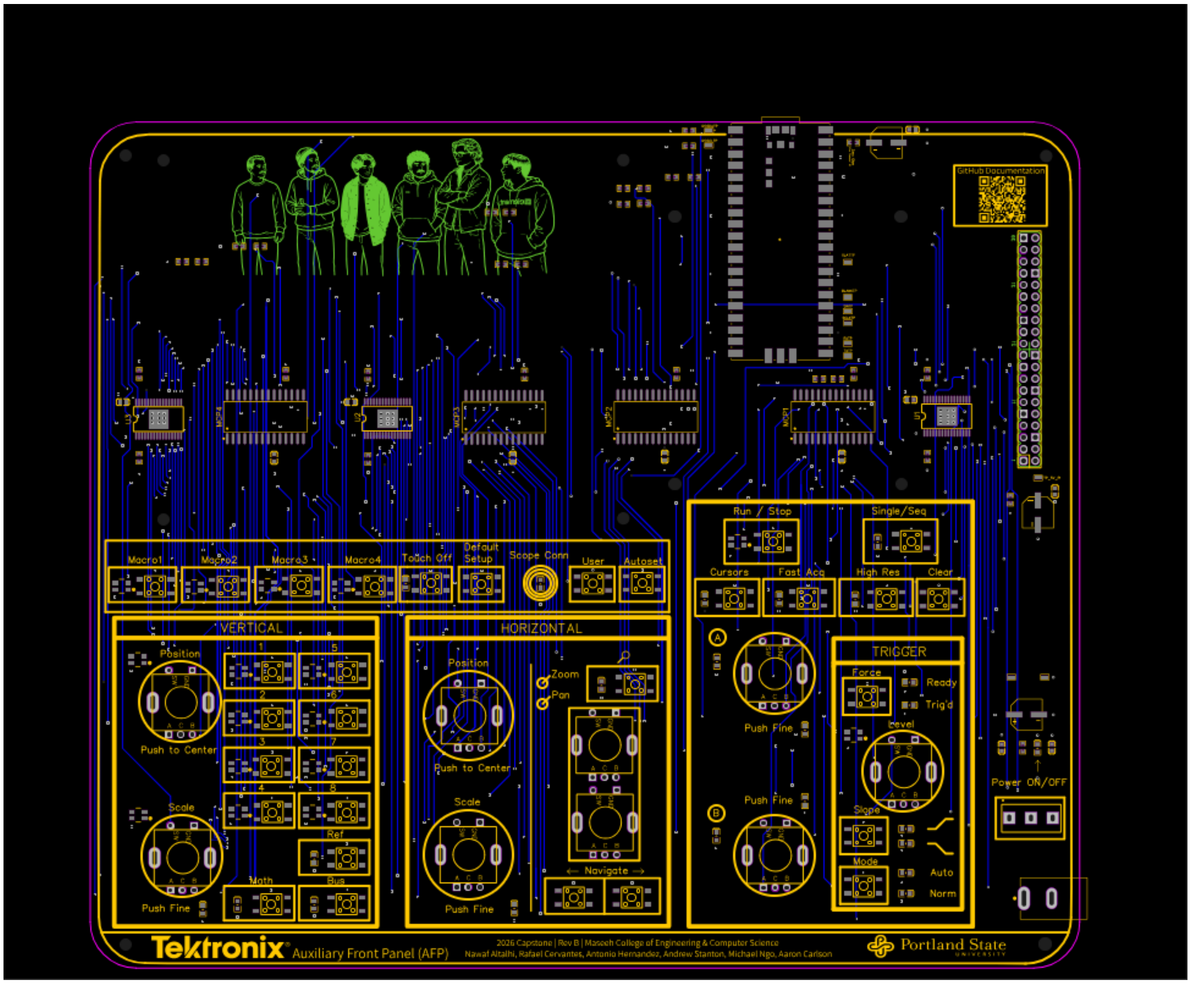


Figure 8.4. L4 Bottom layer — routing

8.3 Routing Strategy

- L2 is used as the main GND plane. The goal was to keep this layer as continuous as possible to provide a clean return path for the digital signals on the top and bottom layers.
- L3 is used for power distribution, mainly for the +5 V and +3.3 V rails. The +5 V rail supports the Pi 4 connection and LED supply, while the +3.3 V rail supports the Pico logic, MCP23017 expanders, TLC5947 logic supply, and pull-up resistors.
- The I²C bus is routed from the Pico to the four MCP23017 I/O expanders. The I²C pull-up resistors R3 and R4 are placed near the Pico side of the bus.
- The TLC5947 control signals, including SCLK, SIN, XLAT, and BLANK, are routed from the Pico to the LED driver chain. Rev B does not include small series resistors on these TLC5947 control lines.

If signal ringing or LED communication issues are observed during bring-up, 33–47 Ω series resistors placed near the Pico outputs should be considered for a future revision.

- The 220 Ω resistors used in Rev B are on the MCP23017 interrupt lines. These resistors are for current limiting and GPIO protection, not for TLC5947 signal termination.

- The LED current path starts from the +5 V supply, passes through the LED anodes, and is sunk by the TLC5947 outputs to ground. Because many LED channels can be active at the same time, the TLC5947 area and its ground connection are important for current return and thermal performance.

- Signal routing was organized around the front-panel layout. Controls were grouped by oscilloscope function, such as acquisition, trigger, vertical, horizontal, macro, and system controls. This helped keep the routing easier to follow and made the PCB more usable as a physical front panel.

8.4 Trace Width Targets

The main routing priority was to keep the higher-current +5 V paths wider than the low-current signal traces. The +5 V rail supplies the Raspberry Pi 4 and the TLC5947 LED system, so these paths need more copper width than the I²C, SPI, UART, button, and encoder signals. The digital control signals carry very little current, so standard signal trace widths are acceptable for those nets.

8.5 Mechanical Mounting

- The PCB includes multiple distributed mounting holes for mechanical support and attachment to a rear plate, enclosure, or stand.

- The mounting holes are designed for M2.5 hardware. A 2.7 mm clearance hole is used to provide enough tolerance for M2.5 screws and standoffs.

- Four mounting holes are placed to match the Raspberry Pi 4 mounting pattern. The Pi 4 is mounted on the rear side of the AFP PCB using M2.5 standoffs and connects through the 2×20 GPIO header area.

- Typical standoff height is expected to be around 8–11 mm, depending on the final connector height and mechanical clearance.

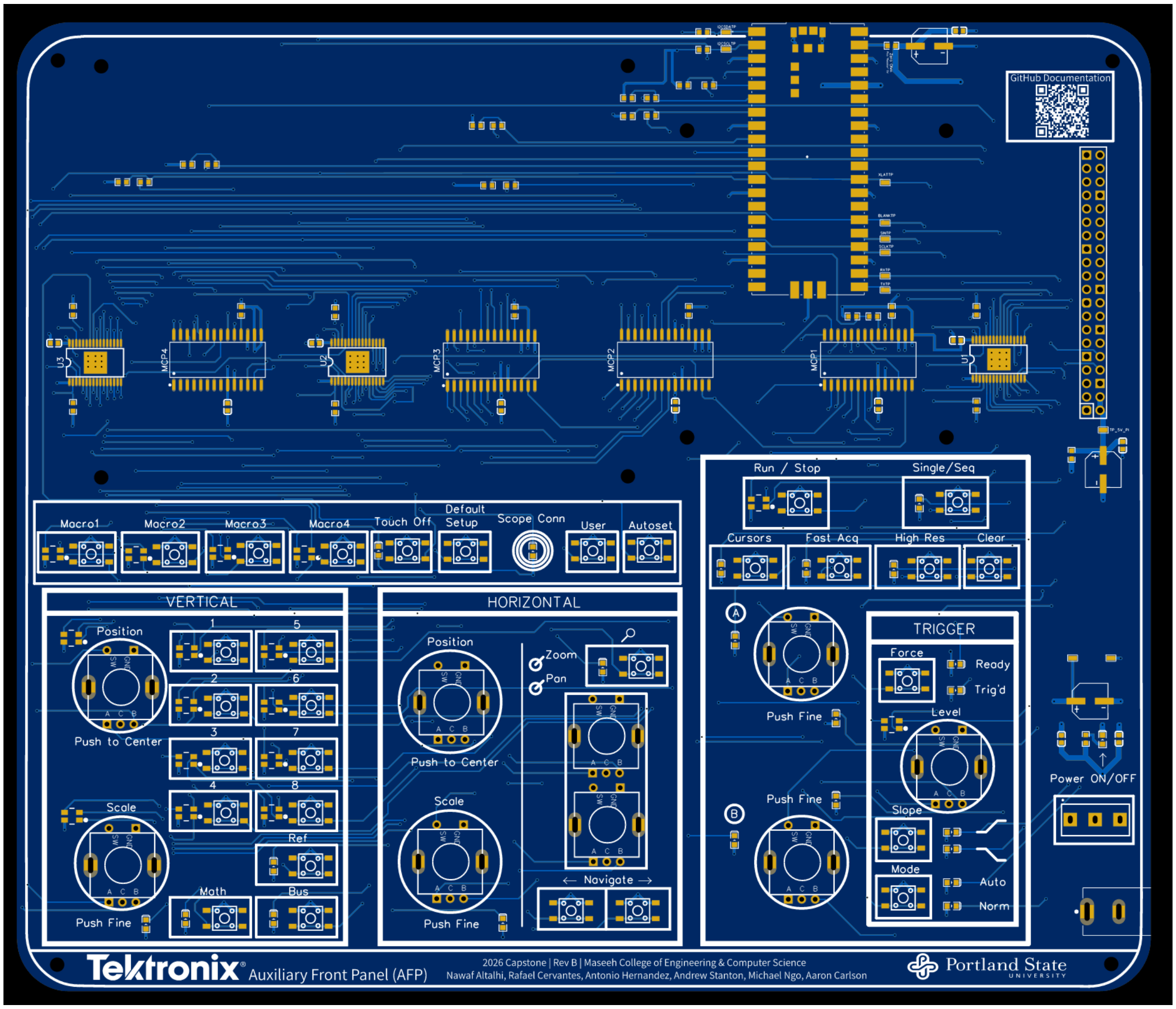


Figure 8.5. 2D bottom-side view

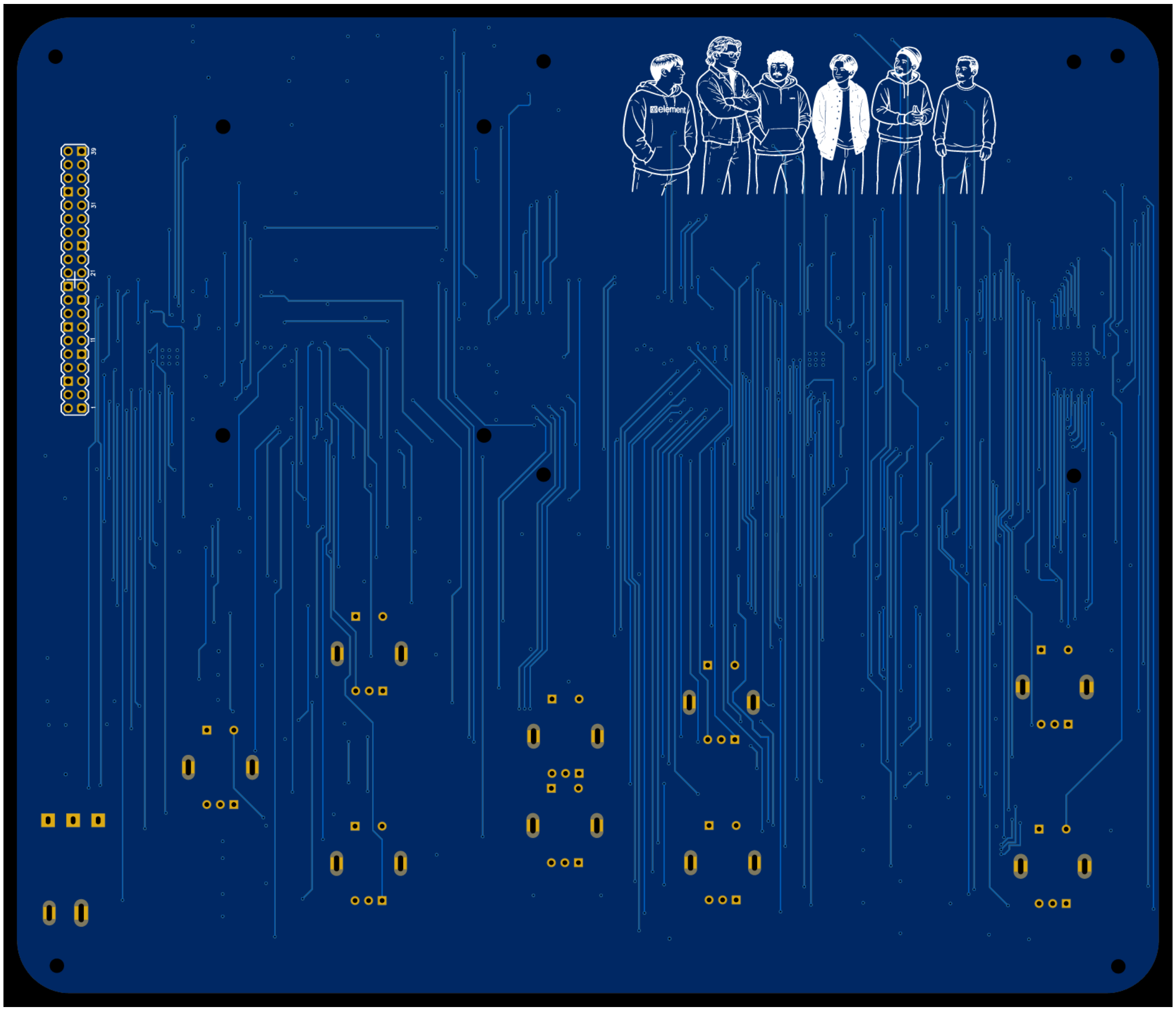


Figure 8.6. 2D bottom-side view

8.6 Final Layout Review Notes

- The TLC5947 thermal pad areas and via connections were included to support heat transfer into the ground plane.
- The +5 V routing was reviewed for the main current paths, especially the Raspberry Pi 4 supply path and the LEDs supply.
- The ground plane was kept as continuous as possible across the entire PCB to provide a stable reference, reduce noise, and support clean return paths for the Pico, MCP23017 expanders, TLC5947 LED drivers, UART, I²C, SPI, and LED current paths.
- Mounting holes were included for the Raspberry Pi 4, the display/screen area, and the board corners for general back-plate support. This gives the finished PCB better mechanical support and provides attachment points for a future enclosure or stand.

9. Assembly

9.1 Component Counts (approximate)

Category	Qty
Resistors (0603)	31
Capacitors (0603 ceramic decoupling)	11
Bulk capacitors (electrolytic)	3
LEDs single-color (SMD)	23
LEDs RGB (SMD common-anode)	16
Pushbuttons (SMD)	31
Rotary encoders (5-pin EC11)	9
ICs	8 (4 × MCP23017 + 3 × TLC5947 + 1 Pico module)
Connectors	2 (J1 barrel jack, GPIO1 2×20 header)
Switches	1 (S2 slide)

Total: 135 pieces per board. The ceramic capacitors are used for local decoupling, while the electrolytic capacitors are used for bulk capacitance on the main power rails.

9.2 Pre-Assembly Checks

- Visually inspect the bare PCB under magnification for solder mask defects, exposed copper, missing labels, or manufacturing defects.
- Multimeter continuity: +5 V to GND must NOT be short. Verify +5 V continuity across the rail. Verify GND continuity.
- Verify part counts and values against BOM (Appendix C)

9.3 SMD Reflow

Stencil + paste, place, reflow in EPL Lab oven. Standard profile:

Stage	Temp	Time
Preheat ramp	25 → 150 °C	60–90 s, ≤3 °C/s
Soak	150–180 °C	60–120 s
Reflow peak	220–245 °C	30–60 s above liquidus
Cooling	Ramp down	4–6 °C/s

Exact profile depends on paste type and oven. Confirm against EPL Lab oven profile and the paste datasheet.

9.4 Through-Hole (after SMD)

- Order: lowest to tallest. Encoders ⟶ barrel jack J1 ⟶ slide switch S2 ⟶ GPIO header
- Tack one corner pin, verify alignment, finish remaining pins
- Trim leads to 1–2 mm above board after soldering
- Iron 350–380 °C leaded / 380–410 °C lead-free

9.5 Post-Solder

- Inspect every joint under magnification — cold joints, tombstones, bridges, insufficient solder
- Continuity recheck: +5 V to GND no-short, +3.3 V to GND no-short
- Clean with IPA (90% +) and lint-free swab. Pay attention to TLC thermal pads where flux pools
- Dry for 30 min room temp or 5 min warm air before power-on

9.6 Mechanical (Pi 4 Attach)

- Install M2.5 standoffs (8–11 mm) through the four Pi-pattern mounting holes from AFP top side
- Lower Pi 4 onto standoffs, aligning 2×20 female header with AFP male header. Press fully seated.
- Verify pin alignment visually before securing standoffs
- Secure with 4 × M2.5 screws

Figure 9.1. Assembled PCB, top side view

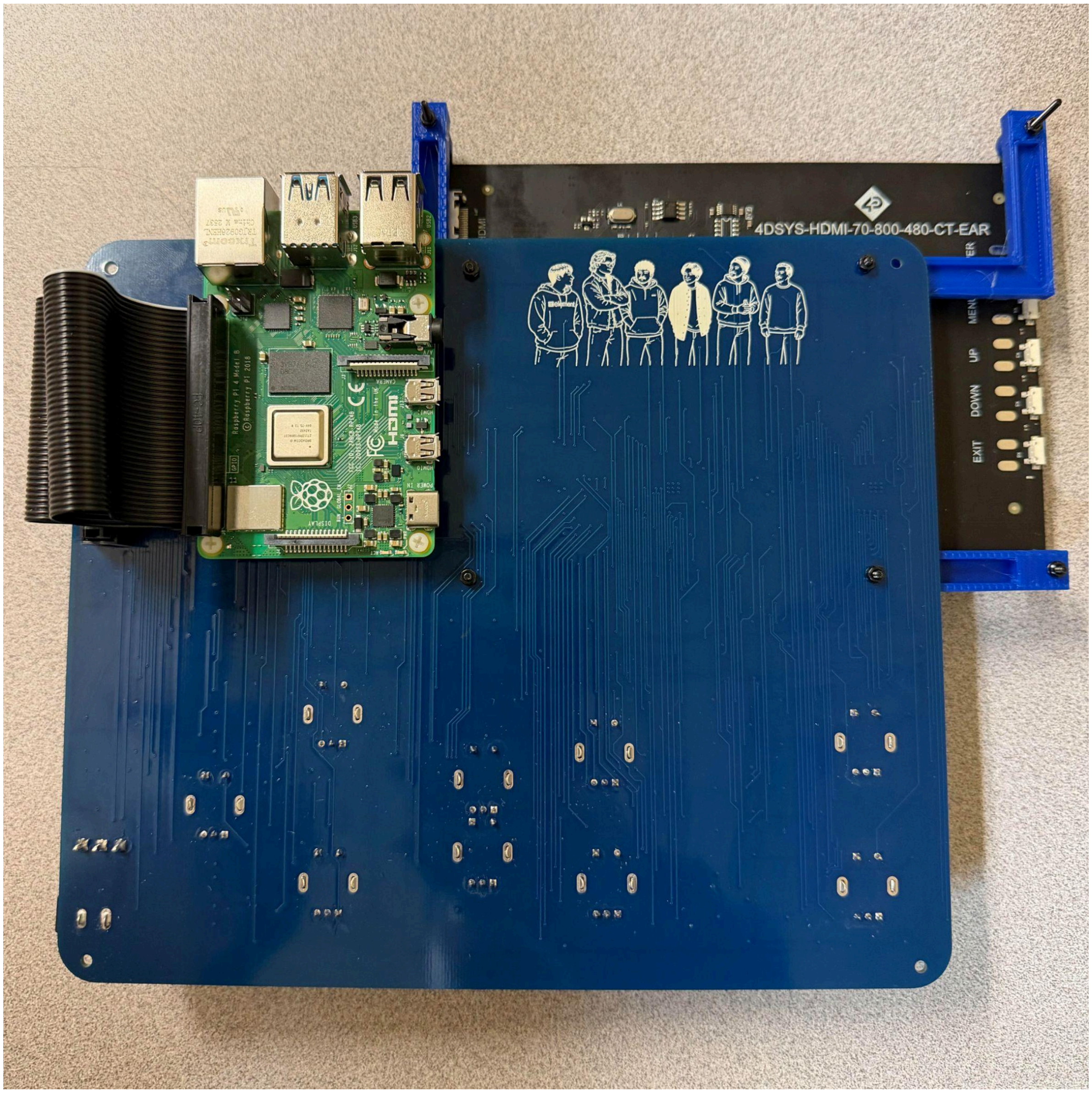


Figure 9.2. Assembled PCB, bottom side with Raspberry Pi 4 attached via M2.5 standoffs and 2×20 header

9.7 Known Assembly Issues (Rev B build)

- SMD placement must be checked carefully before reflow, especially for the small tactile buttons and 0603 passive components.
- Some small SMD parts may shift during reflow if the solder paste amount is uneven or if the part is not centered on the pads before entering the oven.
- 0603 capacitor tombstoning may occur during oven reflow if one pad heats or wets faster than the other. Any affected parts should be corrected by hand after reflow.
- Flux residue can collect around the TLC5947 thermal pad areas and should be cleaned thoroughly after reflow.
- Through-hole components such as the slide switch, barrel jack, rotary encoders, and GPIO header should be soldered after the SMD reflow step.
- The slide switch and barrel jack require careful hand soldering because their terminals have larger metal mass and need enough heat for proper wetting.

10. Bring-Up and Test

10.1 First Power-On Procedure

The first power-on procedure used for HW-001 was performed with the assembled PCB inspected against the BOM before power was applied. A 5 V barrel-plug power supply was connected to the board, and no USB connections were initially used to avoid creating a dual-source power condition.

- Perform a final visual inspection of the assembled PCB under magnification. Check for solder bridges, cold joints, damaged parts, incorrect polarity, and misaligned components.
- Use a multimeter to verify that there is no short circuit between the main power rail and GND before applying power.
- Connect the 5 V barrel-plug power supply to the PCB through the barrel jack.
- Keep USB disconnected during the first power-on check to avoid powering the board from two sources at the same time.
- Turn the power switch ON and verify that the power LED turns on.
- Measure the 5 V rail and 3.3 V rail at the available test points or rail locations. The expected ranges are 4.75–5.25 V for the 5 V rail and 3.15–3.45 V for the 3.3 V rail.
- Press the front-panel buttons and switches individually to confirm that inputs register correctly through the hardware/firmware test setup.
- Run the LED test firmware, if loaded, and verify that the LED indicators illuminate correctly without abnormal behavior.
- Verify Raspberry Pi Pico communication through GPIO/UART response.
- Turn the power switch OFF and confirm that the power LED turns off and the voltage rails discharge normally.
- Disconnect power, clean the PCB if needed, and perform a final inspection.

This procedure was completed on Rev A as HW-001 on 03/31/2026 and passed all listed steps. The same procedure should be repeated on Rev B before continuing with MCP, TLC, and full system testing.

10.2 Validation Stages

Stage	Test	Rev A	Rev B
1. Power & rails	HW-001 (Nawaf)	PASSED	PASSED
2. MCP I/O	MCP-001 (Michael)	PASSED	PASSED
3. TLC LEDs	TLC-001 (Aaron)	PASSED	PASSED
4. End-to-end with scope	Full system test	Verified during practice demo; all hardware functioning	Verified during practice demo; all hardware functioning

10.3 HW-001 — Power-On Verification

HW-001 verifies basic power-up behavior, rail voltages, switch operation, LED response, button input response, Pico communication, and shutdown behavior. Rev A passed HW-001 on 03/31/2026, and Rev B passed HW-001 on 05/21/2026. The measured rails were within the expected ranges of 4.75–5.25 V for the 5 V rail and 3.15–3.45 V for the 3.3 V rail.

Must be re-run on Rev B before any subsequent stage.

10.4 MCP-001 — Button and Encoder Verification

MCP-001 verifies the button and encoder input system through the Raspberry Pi Pico. The test script scans the I²C bus and confirms that all four MCP23017 expanders are detected at addresses 0x20–0x23. It also checks that each button, encoder rotation signal, and encoder push switch produces the expected input event based on the Appendix A mapping.

This test confirms:

- Four MCP23017 devices are detected on the I²C bus.
- MCP input reads return without communication errors.
- Buttons and encoder push switches register correctly.
- Rotary encoders produce the correct direction response.
- Input events match the schematic-derived MCP23017 allocation.

10.5 TLC-001 — LED Chain Verification

TLC-001 verifies the TLC5947 LED driver chain using the Raspberry Pi Pico. The test script walks through LED channels 0–70 and confirms that each assigned LED turns on when its channel is addressed.

This test confirms:

- The TLC5947 serial interface is working.
- The three TLC5947 drivers are in the correct daisy-chain order.
- Each assigned LED channel responds correctly.
- RGB LEDs follow the expected B/G/R channel order from Appendix B.
- The unused channels remain available for future indicators.

10.6 Test Point Quick Reference

TP	Signal	What to look for
TP1	+5 V at Pi 4 header	4.75–5.25 V — confirms PCB powering Pi 4
TP2	+5 V at barrel jack (before S2)	4.75–5.25 V — confirms adapter delivering
TP3	GND	Reference
TP4	I²C SCL	Clock activity during MCP23017 communication; pulled to 3.3 V when idle
TP5	I²C SDA	Data activity, pulled to 3.3 V idle
TP6	PICO_TLC_SIN	Serial data bursts during LED frame send
TP7	PICO_TLC_BLANK	LOW during normal op (LEDs enabled)
TP8	PICO_TLC_XLAT	Pulses once per frame
TP9	UART_TX_PICO (Pico→Pi)	Activity on button/encoder events
TP10	UART_RX_PICO (Pi→Pico)	Activity on LED state updates from Pi
TP11	PICO_TLC_SCLK	Clock activity during TLC5947 LED updates

10.7 Common Bring-Up Failures

Issue	Likely Cause / Check
All LEDs are dark on Rev B	Firmware may still be using the Rev A BLANK pin assignment. Update BLANK from GP15 to GP20.
MCP bus scan finds 0 devices	Check that the 3.3 V rail is present and that the Pico is running. Also verify R16 and measure 3.3 V at an MCP23017 VDD pin.
MCP scan works but false button presses appear	MCP23017 internal pull-ups may not be enabled. Enable GPPU for all used button and encoder inputs.
UART communication is silent	TX/RX may be swapped or not active. Probe TP9 for Pico-to-Pi activity and TP10 for Pi-to-Pico activity.
Raspberry Pi 4 does not boot	Check that R13 is populated, the GPIO header is seated correctly, and TP1 reads 5 V. Avoid powering the Pi 4 from USB-C and the PCB at the same time.
Pico appears as RPI-RP2 instead of running firmware	Pico may be stuck in BOOTSEL mode or firmware may not be loaded. Check the BOOTSEL area and reload firmware if needed.

11. Rev A → Rev B Changelog

Rev B is an improved revision of the Rev A prototype, not a full re-architecture. The main system architecture remained the same, including the Raspberry Pi 4 and Raspberry Pi Pico controller split, the MCP23017 input expansion, the TLC5947 LED driver chain, and the PCB-as-front-panel approach. Rev A verified the main hardware concept, and Rev B improved the layout, mechanical usability, labeling, display support, and final prototype presentation. Rev B was assembled, tested, and verified successfully as the final hardware revision.

11.1 Electrical Changes

Item	Rev A	Rev B	Reason / Impact
MCP interrupt routing	Not fully implemented in the PCB layout	Dedicated MCP interrupt lines routed to Pico GPIOs	Improves input-response handling and gives firmware the option to use interrupt-driven button and encoder reads instead of relying only on polling.
Touch Off LED	Missing	Added	Restores indicator parity

Item	Rev A	Rev B	Reason / Impact
TLC BLANK pin	Routed to the earlier Pico pin assignment	Updated to the Rev B Pico pin assignment, GP20	Firmware-affecting. Test scripts must update.

11.2 Layout Changes

Item	Rev A	Rev B	Reason
Raspberry Pi Pico placement	USB/programming access	Moved to the board edge	Makes firmware loading, debugging, and USB access easier during bring-up.
Barrel jack placement	Less convenient cable location	Moved to the right edge of the board	Cleaner bench cable layout
Project information on PCB	Limited board identification	Added project title, team information, and QR documentation access	Improves the board's professional appearance and makes the documentation easier to access.
Test points	Smaller or less convenient probing areas	Test-point pads enlarged and labeled	Makes bring-up, measurement, and debugging easier.
Mounting support	Limited mechanical support features	Added mounting holes for enclosure/back-plate support	Makes the PCB easier to secure in enclosure, stand, or back plate.
Board edges	Sharp rectangular corners	Rounded board corners	Improves handling, removes sharp corners, and gives the prototype a more finished appearance.

11.3 Mechanical Changes

- Mounting holes were added for Raspberry Pi 4 support, display/screen support, board-corner support, and future enclosure or back-plate attachment.
- Board corners were rounded to remove sharp edges and give the Rev B prototype a cleaner finished appearance.
- Encoder retaining-tab holes were adjusted to improve fit and reduce mechanical stress during assembly.
- Mechanical support was improved so the PCB can better function as both the circuit board and the physical front-panel surface.

11.4 Silkscreen Changes

- Front-panel labels were reviewed and corrected for Rev B to make the control functions easier to identify.
- The “Signal/Seq” label was corrected to “Single/Seq.”
- Test points were labeled with their signal names to make bring-up and debugging easier.
- R13 and R16 were labeled with their power-link destinations.
- Zoom and Pan labels were separated to improve readability.
- Project title, team information, PSU logo, and QR documentation access were added to the PCB.
- Section outlines and label placement were adjusted to make the PCB-as-front-panel presentation cleaner and more professional.

11.5 Considered but Not Changed

Some design improvements were considered during the Rev B update but were not added because Rev B was focused on completing a reliable working prototype within the capstone schedule.

- Additional input protection, such as a fuse, TVS diode, or reverse-polarity protection, was considered but left for a future revision.
- A dedicated 3.3 V regulator was considered, but the Rev B design continued using the Raspberry Pi Pico 3V3OUT rail for MCP23017 logic, TLC5947 logic, and pull-ups.
- Power-path protection for USB and PCB power sources was considered, but Rev B uses the documented operating procedure of avoiding dual 5 V power sources during programming and setup.
- A separate top front-panel overlay was considered, but Rev B keeps the PCB-as-front-panel approach. A rear stand/enclosure support was created to hold and support the prototype.

11.6 Unchanged (Carried Forward)

- Raspberry Pi 4 and Raspberry Pi Pico controller split with UART communication between them.
- Four MCP23017 I/O expanders on the shared I^2C bus for button and encoder inputs.
- Three TLC5947 LED drivers in a daisy-chain configuration for front-panel LED control.
- Single 5 V barrel-jack input through the main power switch.
- Raspberry Pi Pico 3V3OUT rail used for MCP23017 logic, TLC5947 logic, and pull-ups.
- Active-LOW button and encoder input convention using MCP23017 internal pull-ups.
- PCB-as-front-panel design approach with controls, labels, and LEDs mounted directly on the user-facing PCB.
- Four-layer PCB stackup using Signal / GND / PWR / Signal.

12. Errata and Rev C Recommendations

Rev B was assembled, tested, and verified as the final hardware prototype. The following items are recommended only for a future Rev C if the design is continued beyond the capstone prototype or used outside a controlled lab environment.

12.1 Rev C Priorities

Priority	Item	Reason
High	Add input protection on the 5 V power input	A polyfuse, TVS diode, and reverse-polarity protection would make the board more robust for repeated use outside a controlled lab setting.
High	Add power-path protection for USB and PCB power sources	Ideal-diode or power-path circuitry would reduce the risk of backfeeding when the Pico or Raspberry Pi 4 is connected by USB during programming or setup.
Medium	Add ESD protection on user-facing controls	TVS diode arrays on button and encoder lines would improve protection against static discharge from repeated user interaction.
Low	Add a dedicated 3.3 V regulator	A separate 3.3 V rail would reduce dependence on the Pico 3V3OUT pin and make bench debugging easier.

13. Project Strengths and Lessons Learned

This section summarizes the main hardware strengths of the AFP Rev B prototype and the lessons learned during the Rev A to Rev B development process. The goal is to document what worked well, what design choices were successful, and what could be improved in a future revision. .

13.1 Things That Went Right

Controller split

The Raspberry Pi 4 and Raspberry Pi Pico split worked well for this project. The Pi 4 handles the higher-level software, display, macro storage, and oscilloscope communication, while the Pico handles the real-time hardware I/O for buttons, encoders, and LEDs. This made the hardware easier to organize and kept the oscilloscope-specific logic on the Pi 4 side.

PCB-as-front-panel approach

Using the PCB as the physical front panel was a successful design choice. It reduced wiring, kept the controls aligned with the silkscreen labels, and made the prototype easier to assemble and present. This approach also helped the board feel closer to a real oscilloscope-style control surface.

Organized control layout

The controls were grouped by oscilloscope function, including acquisition, trigger, vertical, horizontal, macro, and system controls. This made the board easier to understand and improved the usability of the front-panel layout.

I/O expansion and LED driver architecture

The MCP23017 I/O expanders and TLC5947 LED drivers allowed the board to support a large number of buttons, encoders, and LED indicators without using a large number of Pico GPIO pins. This made the design more scalable and easier to route.

Test and bring-up access

The dedicated test points made bring-up and debugging easier. Power rails, I^2C, TLC5947 control signals, and UART lines can be checked without probing directly on small IC pins.

Rev B improvements from Rev A feedback

Rev A helped prove the main hardware concept. Rev B improved the final prototype by updating the Pico access, barrel jack location, display support, mounting support, silkscreen labels, rounded corners, and general board presentation.

13.2 Things We Would Do Differently

Add input protection earlier
For a future revision, the 5 V input should include protection such as a fuse, TVS diode, and reverse-polarity protection. Rev B works for a controlled capstone prototype, but added input protection would make the design more robust for repeated use.

Plan power-source protection earlier
Rev B uses an operating procedure to avoid powering the Raspberry Pi 4 or Pico from two 5 V sources at the same time. In a future revision, ideal-diode or power-path protection could make USB programming and external power use safer and more user-friendly.

Consider a protective front overlay
The PCB-as-front-panel approach worked well, but a clear protective overlay could improve long-term durability and protect the silkscreen if the design is used beyond the prototype stage.

13.3 What This Project Demonstrates

- A PCB-based auxiliary front panel can provide a practical physical control interface for Tektronix MSO oscilloscopes.
- The Raspberry Pi 4 and Raspberry Pi Pico architecture can successfully separate high-level oscilloscope communication from real-time front-panel I/O.
- A 4-layer PCB can support a large number of user controls, LED indicators, and digital interfaces while still functioning as the physical front-panel surface.
- The Rev A prototype provided useful test results, and Rev B used those results to improve layout, assembly, mounting, labeling, and overall usability.
- The final Rev B hardware prototype was assembled, tested, and verified successfully, demonstrating that the main hardware design is complete and functional.

Appendix A. MCP23017 Pin Allocation

Verified from EasyEDA netlist (MCP23017T-E-SO_2026-05-16.net). Pin numbering follows MCP23017 SOIC-28 datasheet convention (GPA0–GPA7 = pins 21–28, GPB0–GPB7 = pins 1–8). All inputs active LOW; firmware must enable internal pull-ups.

A.1 MCP1 (0x20)

Pin	Port.Bit	Signal	Designator
21	A.0	Single/Seq Button	U4
22	A.1	Run/Stop Button	U7
23	A.2	Clear Button	U12
24	A.3	High Res Button	U17
25	A.4	Fast Acq Button	U19
26	A.5	Cursor Button	U18
27	A.6	Force Button	U22
28	A.7	Slope Button	U21
1	B.0	Mode Button	U20
2	B.1	Cursor A enc — Phase A	U15.3
3	B.2	Cursor A enc — Phase B	U15.5
4	B.3	Cursor A enc — Push	U15.1
5	B.4	Cursor B enc — Phase A	U14.3
6	B.5	Cursor B enc — Phase B	U14.5
7	B.6	Cursor B enc — Push	U14.1
8	B.7	Trigger Level enc — Phase A (B/SW on MCP2)	U16.3

A.2 MCP2 (0x21)

Pin	Port.Bit	Signal	Designator
21	A.0	Trigger Level enc — Phase B (A on MCP1)	U16.5
22	A.1	Trigger Level enc — Push	U16.1
23	A.2	Vertical Position enc — Phase A	U11.3
24	A.3	Vertical Position enc — Phase B	U11.5
25	A.4	Vertical Position enc — Push	U11.1

Pin	Port.Bit	Signal	Designator
26	A.5	Vertical Scale enc — Phase A	U6.3
27	A.6	Vertical Scale enc — Phase B	U6.5
28	A.7	Vertical Scale enc — Push	U6.1
1	B.0	Ch1 Button	U25
2	B.1	Ch2 Button	U24
3	B.2	Ch3 Button	U23
4	B.3	Ch4 Button	U28
5	B.4	Ch5 Button	U29
6	B.5	Ch6 Button	U30
7	B.6	Ch7 Button	U31
8	B.7	Ch8 Button	U26

A.3 MCP3 (0x22)

Pin	Port.Bit	Signal	Designator
21	A.0	Math Button	U27
22	A.1	Ref Button	U33
23	A.2	Bus Button	U34
24	A.3	Horizontal Position enc — Phase A	U13.3
25	A.4	Horizontal Position enc — Phase B	U13.5
26	A.5	Horizontal Position enc — Push	U13.1
27	A.6	Horizontal Scale enc — Phase A	U10.3
28	A.7	Horizontal Scale enc — Phase B	U10.5
1	B.0	Horizontal Scale enc — Push	U10.1
2	B.1	Zoom enc — Phase A (no push)	U8.3
3	B.2	Zoom enc — Phase B	U8.5
4	B.3	Pan enc — Phase A (no push)	U9.3
5	B.4	Pan enc — Phase B	U9.5
6	B.5	Navigate Left Button	U35
7	B.6	Navigate Right Button	U36
8	B.7	Zoom Button (push)	U37

A.4 MCP4 (0x23)

Pin	Port.Bit	Signal	Designator
21	A.0	Macro 1 Button	U38
22	A.1	Macro 2 Button	U39
23	A.2	Macro 3 Button	U40
24	A.3	Macro 4 Button	U32
25	A.4	Touch Off Button	U43
26	A.5	User Button	U42
27	A.6	Default Setup Button	U41
28	A.7	Autoset Button	U44
1–8	B.0–7	(unused, available for expansion)	—

Appendix B. TLC5947 Channel Allocation

Verified from netlist. TLC5947 pin numbering: GND=1, BLANK=2, SCLK=3, SIN=4, OUT0=5...OUT23=28, SOUT=29, XLAT=30, IREF=31, VCC=32, thermal pad=33. Channel number TLCn_K maps 1:1 to OUT_K.

RGB LEDs: pin 2 = common anode (+5 V). Pin 4 = Blue cathode, pin 3 = Green cathode, pin 1 = Red cathode. In the channel sequence for each RGB LED, lowest channel = B, middle = G, highest = R.

B.1 TLC1 (U1)

Ch	Pin	LED.pin	Function
0	5	LED14.4	Trigger Level ring [B]
1	6	LED14.3	Trigger Level ring [G]
2	7	LED14.1	Trigger Level ring [R]
3	8	LED4.4	Run/Stop [B]
4	9	LED4.3	Run/Stop [G]
5	10	LED4.1	Run/Stop [R]
6	11	LED3	Single/Seq indicator
7	12	LED5	High Res indicator
8	13	LED6	Fast Acq indicator
9	14	LED7	Cursors (white)
10	15	LED8	Cursor A ring (white)
11	16	LED9	Cursor A Fine (white)
12	17	LED10	Cursor B ring (white)
13	18	LED11	Cursor B Fine (white)
14	19	LED12	Force Ready (white)
15	20	LED13	Trig'd
16	21	LED15	Slope Up
17	22	LED16	Slope Down
18	23	LED17	Mode Auto
19	24	LED18	Mode Norm
20	25	LED21	Vertical Scale Fine (white)
21	26	LED30	Math (orange)
22	27	LED31	Ref (white)

Ch	Pin	LED.pin	Function
23	28	LED32	Bus (purple)

B.2 TLC2 (U2)

Ch	Pin	LED.pin	Function
0	5	LED19.4	Vertical Position [B]
1	6	LED19.3	Vertical Position [G]
2	7	LED19.1	Vertical Position [R]
3	8	LED20.4	Vertical Scale [B]
4	9	LED20.3	Vertical Scale [G]
5	10	LED20.1	Vertical Scale [R]
6	11	LED22.4	Ch1 [B]
7	12	LED22.3	Ch1 [G]
8	13	LED22.1	Ch1 [R]
9	14	LED23.4	Ch2 [B]
10	15	LED23.3	Ch2 [G]
11	16	LED23.1	Ch2 [R]
12	17	LED24.4	Ch3 [B]
13	18	LED24.3	Ch3 [G]
14	19	LED24.1	Ch3 [R]
15	20	LED25.4	Ch4 [B]
16	21	LED25.3	Ch4 [G]
17	22	LED25.1	Ch4 [R]
18	23	LED26.4	Ch5 [B]
19	24	LED26.3	Ch5 [G]
20	25	LED26.1	Ch5 [R]
21	26	LED27.4	Ch6 [B]
22	27	LED27.3	Ch6 [G]
23	28	LED27.1	Ch6 [R]

B.3 TLC3 (U3)

Ch	Pin	LED.pin	Function
0	5	LED28.4	Ch7 [B]
1	6	LED28.3	Ch7 [G]
2	7	LED28.1	Ch7 [R]
3	8	LED29.4	Ch8 [B]
4	9	LED29.3	Ch8 [G]
5	10	LED29.1	Ch8 [R]
6	11	LED33	Zoom indicator (white)
7	12	LED38.4	Macro 1 [B]
8	13	LED38.3	Macro 1 [G]
9	14	LED38.1	Macro 1 [R]
10	15	LED37.4	Macro 2 [B]
11	16	LED37.3	Macro 2 [G]
12	17	LED37.1	Macro 2 [R]
13	18	LED36.4	Macro 3 [B]
14	19	LED36.3	Macro 3 [G]
15	20	LED36.1	Macro 3 [R]
16	21	LED35.4	Macro 4 [B]
17	22	LED35.3	Macro 4 [G]
18	23	LED35.1	Macro 4 [R]
19	24	LED34	Scope Connected (green)
20	25	LED1	Horizontal Scale Fine (white)
21	26	LED39	Touch Off (white) — Rev B addition
22	27	—	Unused
23	28	—	Unused

Appendix C. Net Naming Convention

Used throughout schematic and netlist. Helps readers trace signals quickly.

- I2C_SDA, I2C_SCL — shared I^2C bus to all MCPs
- MCPn_INTA, MCPn_INTB — interrupt outputs from MCP n (n = 1–4)
- MCPnA_K, MCPnB_K — GPIO bit K of MCP n Port A or B (K = 0–7)
- PICO_TLC_SCLK, PICO_TLC_SIN, PICO_TLC_BLANK, PICO_TLC_XLAT — TLC chain control signals from Pico
- SOUT_TLC2, SOUT_TLC3 — daisy-chain data outputs (TLC1.SOUT → TLC2.SIN, TLC2.SOUT → TLC3.SIN)
- TLCn_K — OUT pin K of TLC n; net then connects to the corresponding LED cathode
- UART_TX_PICO, UART_RX_PICO — named for the source (Pico). UART_TX_PICO is driven by Pico, and lands on Pi RX. UART_RX_PICO is driven by Pi, and lands on Pico RX.
- +5V, +3.3V, GND — power rails

Final Note

This AFP Hardware Reference Manual documents the final Rev B hardware prototype, including the system architecture, power design, processor connections, input expansion, LED driver system, PCB layout, assembly process, bring-up testing, and Rev A to Rev B improvements.

Rev B was assembled, tested, and verified successfully as the final hardware revision for the capstone project. The related GitHub repository contains the supporting project documentation and design files needed for future reference, review, or continuation of the hardware work.

A QR code is provided with this manual to give direct access to the project documentation and repository resources.